The HHL Algorithm

Assessment of the Harrow–Hassidim–Lloyd algorithm for $A\vec{x} = \vec{b}$

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key**, section by section, with the topic tested by every question (page 3),
2. a **topic-wise map** grouping the questions by theme and a **quick-marking grid** (page 4), and
3. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 5).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count N_c = number of correct answers and N_w = number of questions answered wrongly (or with more than one bubble darkened). Section marks = $3N_c - N_w$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy linear algebra, NumPy state vectors of the HHL circuits in the notebook's convention, and CUDA-Q for the notebook's HHL kernel and the phase-estimation circuits) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- Linear system $A\vec{x} = \vec{b}$, A a $d \times d$ matrix, $d = 2^{n_b}$. A Hermitian with eigenpairs $A|u_j\rangle = \lambda_j |u_j\rangle$; $|b\rangle = \vec{b}/\|\vec{b}\| = \sum_j \beta_j |u_j\rangle$; $|x\rangle = A^{-1}|b\rangle / \|A^{-1}|b\rangle\|$. Condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (Hermitian A), in general $\kappa = \sigma_{\max}/\sigma_{\min}$. Sparsity s = largest number of non-zero entries in a row.
- Basis states are written with **qubit 0 first = most significant bit**: $|q_0 q_1 \ldots\rangle$, e.g. $|10\rangle = |2\rangle$. $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_Y(\theta) = \left(\begin{smallmatrix}\cos\frac{\theta}{2} & -\sin\frac{\theta}{2}\\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{smallmatrix}\right)$; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. $\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{N}}\sum_y e^{2\pi i\, xy/N}|y\rangle$, $N = 2^n$.
- **Phase estimation (QPE):** n_c clock qubits in $|+\rangle$ control U^{2^j}, then the inverse QFT; an eigenphase $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ gives the outcome $k \approx \varphi\, 2^{n_c}$.
- **HHL (as in `HHL.ipynb`):** registers clock (n_c), system (n_b), ancilla (1), total $n_c + n_b + 1$ qubits. $U = e^{iAt}$, so eigenvalue λ gives clock value $k = \lambda t\, 2^{n_c}/(2\pi)$. $C = 2\pi/(2^{n_c} t)$ is the smallest representable eigenvalue; for clock value k ($\tilde{\lambda}_k = kC$) the ancilla is rotated by $R_Y(\theta_k)$, $\theta_k = 2\arcsin(C/\tilde{\lambda}_k)$. Clock values $k \geq 2^{n_c-1}$ mean the negative value $k - 2^{n_c}$ (two's complement) unless a question says otherwise. The run succeeds when the ancilla is measured as 1 (and the clock as $0\ldots0$); $P_{\text{success}} = \sum_j |\beta_j|^2 C^2/\lambda_j^2$ when every λ_j is exactly representable.
- **Notebook default:** rows of A: $(2.5, -0.5, -1, 0)$, $(-0.5, 2.5, 0, -1)$, $(-1, 0, 2.5, -0.5)$, $(0, -1, -0.5, 2.5)$ (eigenvalues $1, 2, 3, 4$), $\vec{b} = (1, 0, 0, 0)$, $n_c = 4$, $t = \pi/8$, $C = 1$, 7 qubits, 20 000 shots. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).

- log means $\log_2$; $\tilde{O}$ hides polylogarithmic factors. Numerical answers are rounded.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Linear Algebra and Mathematical Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**D**	Solving a 2×2 system	11	**B**	Normalisation and amplitude encoding
2	**C**	Eigenvalues and κ of a symmetric matrix	12	**C**	Solving $A\vec{x} = \vec{b}$ in the eigenbasis
3	**B**	Matrix properties (Hermitian, unitary, singular)	13	**D**	Norms, κ and determinant
4	**D**	Condition number of a diagonal matrix	14	**D**	Hermitian embedding of a non-Hermitian matrix
5	**C**	Eigenvalues of a complex Hermitian matrix	15	**C**	When is e^{iAt} unitary?
6	**A**	Inverse via spectral decomposition	16	**A**	Error amplification by κ
7	**B**	Spectral theorem vs. real eigenvalues	17	**D**	2×2 example: eigenvalues
8	**A**	Condition number of U and of A^{-1}	18	**C**	2×2 example: normalised solution
9	**B**	Sparsity of the 1-D Laplacian	19	**B**	Embedding + padding: qubit count
10	**A**	Spectrum and κ of the Laplacian	20	**A**	κ from singular values

Section B — Quantum Computing Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**C**	Born rule	31	**B**	Global vs. relative phase
22	**A**	State-vector size and memory	32	**A**	Controlled-R_Y and ancilla probability
23	**C**	Gate identity $HZH = X$	33	**C**	Selecting a clock value with X gates
24	**B**	R_Y rotations of $\|0\rangle$	34	**A**	Post-selection on an ancilla
25	**C**	Entanglement test	35	**D**	Inverse of a circuit
26	**D**	Phase kickback	36	**D**	Gate counts of a 4-qubit QFT
27	**D**	Inverse QFT of a phase state	37	**B**	Uniform superposition
28	**B**	QPE with an exact 3-bit phase	38	**D**	Product states, reduced states
29	**A**	QPE with a non-representable phase	39	**B**	QPE trace: phase state
30	**C**	Unitarity vs. measurement	40	**A**	QPE trace: outcome

Section C — HHL Algorithm Fundamentals

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**C**	What HHL outputs	51	**C**	Negative eigenvalues (two's complement)
42	**D**	State after phase estimation	52	**A**	Code: X gates of the inversion
43	**C**	Eigenvalue $\rightarrow$ clock bit string	53	**B**	Norm recovery $\vec{x} = \|\vec{b}\|\,\vec{a}/C$
44	**B**	Constant C and rotation angle	54	**A**	Default 4×4 solution
45	**D**	Ancilla amplitude C/λ	55	**D**	Algorithm step $\leftrightarrow$ CUDA-Q code
46	**A**	Roles of the registers	56	**B**	Preparing $\|b\rangle$
47	**D**	Success probability	57	**B**	Role of the inverse QFT
48	**B**	Qubit count $n_b + n_c + 1$	58	**A**	$\|b\rangle$ an eigenvector
49	**C**	Why uncompute?	59	**D**	2×2 HHL trace: after QPE
50	**A**	Choice of C	60	**C**	2×2 HHL trace: output

Section D — Advanced HHL and Implementation

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**C**	Complexity of HHL	71	**C**	Recorded P_{success}: shot statistics
62	**B**	Repetitions and amplitude amplification	72	**A**	Recorded read-out of $\lvert x_0\rvert^2$
63	**C**	Quantum linear-system algorithms	73	**B**	Circuit resources vs. n_c
64	**B**	Read-out cost of all x_i	74	**D**	Error sources and their effects
65	**D**	"Exponential speed-up" claims	75	**B**	Useful outputs of HHL
66	**A**	Too few clock qubits: rescaled t	76	**B**	Dequantisation
67	**D**	Clock qubits for precision ϵ	77	**A**	κ^2 run-time dependence
68	**D**	Memory of a large HHL simulation	78	**D**	Solution inside the embedded system
69	**A**	Trotter steps for e^{iAt}	79	**A**	Recorded run: P_{success} exactly
70	**C**	What the notebook implements	80	**C**	Recorded run: $\lVert\vec{x}\rVert$ from P_{success}

Topic-wise map of the questions

Every question appears under at least one theme; several appear under two.

Theme	Question numbers
Linear systems and their classical solution	1, 6, 12, 16, 18, 54
Eigenvalues, Hermitian matrices and spectral decomposition	2, 3, 5, 6, 7, 10, 12, 17
Condition number, norms and sparsity	4, 8, 9, 10, 13, 16, 20, 77
Hermitian embedding, padding and unitarity of e^{iAt}	14, 15, 19, 78
Normalisation, amplitude encoding and state preparation	11, 22, 37, 56
Qubits, measurement, phases and entanglement	21, 23, 25, 30, 31, 38
Gates, controlled operations and circuits	24, 26, 32, 33, 35, 36, 52
QFT and quantum phase estimation	27, 28, 29, 36, 39, 40, 42, 43, 57, 66, 67
Eigenvalue inversion, the constant C and the ancilla	44, 45, 50, 51, 52, 59, 60
Post-selection, success probability and uncomputation	34, 47, 49, 53, 58, 60, 62
HHL registers, circuit structure and resources	46, 48, 55, 68, 73
Complexity, read-out and advanced algorithms	41, 61, 63, 64, 65, 69, 75, 76, 77
Noise, errors and limitations	66, 70, 74, 76
Interpreting recorded results of `HHL.ipynb`	54, 70, 71, 72, 79, 80

Quick-marking grid

1 **D**	2 **C**	3 **B**	4 **D**	5 **C**	6 **A**	7 **B**	8 **A**	9 **B**	10 **A**
11 **B**	12 **C**	13 **D**	14 **D**	15 **C**	16 **A**	17 **D**	18 **C**	19 **B**	20 **A**
21 **C**	22 **A**	23 **C**	24 **B**	25 **C**	26 **D**	27 **D**	28 **B**	29 **A**	30 **C**
31 **B**	32 **A**	33 **C**	34 **A**	35 **D**	36 **D**	37 **B**	38 **D**	39 **B**	40 **A**
41 **C**	42 **D**	43 **C**	44 **B**	45 **D**	46 **A**	47 **D**	48 **B**	49 **C**	50 **A**
51 **C**	52 **A**	53 **B**	54 **A**	55 **D**	56 **B**	57 **B**	58 **A**	59 **D**	60 **C**
61 **C**	62 **B**	63 **C**	64 **B**	65 **D**	66 **A**	67 **D**	68 **D**	69 **A**	70 **C**
71 **C**	72 **A**	73 **B**	74 **D**	75 **B**	76 **B**	77 **A**	78 **D**	79 **A**	80 **C**

Answer Bank — questions with detailed solutions

SECTION A · Linear Algebra and Mathematical Foundations Q.1–20

Q.1 SINGLE CORRECT · Solving a 2×2 system

The solution of the linear system $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is

(A) $(x_1, x_2) = (-0.2,\ 0.6)$
(B) $(x_1, x_2) = (0.6,\ 0.2)$
(C) $(x_1, x_2) = (1,\ 0.\overline{3})$
(D) ✓ $(x_1, x_2) = (0.2,\ 0.6)$

Answer: (D) Detailed solution

Correct answer. **(D)** $(x_1, x_2) = (0.2,\ 0.6)$
Concepts required. A square system $A\vec{x} = \vec{b}$ has a unique solution iff $\det A \neq 0$; for 2×2 matrices the inverse has a closed form.
Relevant formulae. $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$; $\vec{x} = A^{-1}\vec{b}$.
Step-by-step explanation. $\det A = 2 \cdot 3 - 1 \cdot 1 = 5 \neq 0$, so the system has a unique solution. For a 2×2 matrix

$$A^{-1} = \frac{1}{\det A}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \vec{x} = A^{-1}\vec{b} = \frac{1}{5}\begin{pmatrix} 3 \cdot 1 - 1 \cdot 2 \\ -1 \cdot 1 + 2 \cdot 2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.6 \end{pmatrix}.$$

Check: $2(0.2) + 0.6 = 1$ and $0.2 + 3(0.6) = 2$. (Gaussian elimination gives the same: $R_2 \leftarrow R_2 - \frac{1}{2}R_1$ gives $2.5x_2 = 1.5$.)
Why the other options are incorrect.
(A) $(-0.2, 0.6)$ has a sign error in the first component: $2(-0.2) + 0.6 = 0.2 \neq 1$.
(B) $(0.6, 0.2)$ swaps the two components; substituting gives $2(0.6) + 0.2 = 1.4 \neq 1$.
(C) $(1, 0.\overline{3})$ solves each equation separately ($2x_1 = 2$?) and ignores the coupling; $2(1) + \frac{1}{3} \neq 1$.
Key takeaway. Always substitute the candidate back into $A\vec{x} = \vec{b}$: it takes ten seconds and catches every slip.

Q.2 SINGLE CORRECT · Eigenvalues and κ of a symmetric matrix

The eigenvalues and the condition number κ of $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$ are

(A) 3 and 4; $\kappa = 4/3$
(B) 2 and 5; $\kappa = 2.5$
(C) ✓ 1 and 6; $\kappa = 6$
(D) 1 and 6; $\kappa = 7$

Answer: (C) Detailed solution

Correct answer. **(C)** 1 and 6; $\kappa = 6$
Concepts required. Eigenvalues are the roots of $\det(A - \lambda I) = 0$; for a symmetric matrix they are real and $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$.
Relevant formulae. 2×2: $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$; $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$.
Step-by-step explanation. Characteristic polynomial: $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6)$, so $\lambda \in \{1, 6\}$. Eigenvectors: $\lambda = 1$: $4x + 2y = 0 \Rightarrow (1, -2)$; $\lambda = 6$: $-x + 2y = 0 \Rightarrow (2, 1)$ — orthogonal, as they must be for a symmetric matrix. A is symmetric positive definite, so

$$\kappa = \frac{|\lambda|_{\max}}{|\lambda|_{\min}} = \frac{6}{1} = 6.$$

(The diagonal entries $5, 2$ are *not* the eigenvalues, because A is not diagonal.)

Why the other options are incorrect.

(A) 3 and 4 have the right sum 7 but product $12 \neq 6$, so they are not the roots.

(B) 2 and 5 are the diagonal entries, which are eigenvalues only for diagonal (or triangular) matrices; their product $10 \neq \det A = 6$.

(D) $\kappa = 7$ adds the eigenvalues ($\operatorname{tr} A$) instead of dividing the largest by the smallest.

Key takeaway. Check eigenvalues with two invariants: their sum is the trace and their product the determinant.

Q.3 MATCHING LISTS · Matrix properties (Hermitian, unitary, singular)

Match each matrix (List-I) with the description that fits it (List-II).

List-I		List-II	
(P)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	**(1)**	Hermitian and unitary
(Q)	$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$	**(2)**	Hermitian, invertible, not unitary
(R)	$\begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$	**(3)**	Hermitian but singular
(S)	$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$	**(4)**	not diagonalisable

(A) P→2, Q→3, R→1, S→4 **(B)** ✓ P→1, Q→3, R→2, S→4

(C) P→1, Q→4, R→2, S→3 **(D)** P→1, Q→2, R→3, S→4

Answer: (B) Detailed solution

Correct answer. (B) P→1, Q→3, R→2, S→4

Concepts required. Hermitian: $A = A^\dagger$. Unitary: $A^\dagger A = I$. Singular: $\det A = 0$. Diagonalisable: a full set of eigenvectors.

Relevant formulae. $A^\dagger = \bar{A}^T$; $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$; Hermitian ⇒ unitarily diagonalisable.

Step-by-step explanation.

- P $= X$: real symmetric, so Hermitian; $X^\dagger X = X^2 = I$, so unitary (1).
- Q: symmetric (Hermitian) but $\det = 4 - 4 = 0$ (second row = 2× first): singular, eigenvalues 0 and 5 (3).
- R: $R^\dagger$ = conjugate transpose $= \left(\begin{smallmatrix} 1 & \overline{-i} \\ \overline{i} & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & i \\ -i & 2 \end{smallmatrix}\right) = R$, so Hermitian; $\det R = 2 - (i)(-i) = 2 - 1 = 1 \neq 0$, invertible; $R^\dagger R = R^2 \neq I$ (e.g. $(R^2)_{11} = 1 + 1 = 2$), not unitary (2).
- S: both eigenvalues are 0 but $S \neq 0$; the eigenspace of 0 is only 1-dimensional ($\{(a, 0)\}$), so S is not diagonalisable (4).

Only Hermitian, invertible matrices (P and R) are directly usable in HHL.

Why the other options are incorrect.

(A) P→2 calls X "not unitary", but $X^2 = I$; R→1 calls the 2×2 matrix with diagonal $1, 2$ unitary although its first column has norm $\sqrt{2}$.

(C) Q→4: Q is symmetric, hence diagonalisable (eigenvalues 0 and 5); S→3: S is not Hermitian ($S^\dagger \neq S$).

(D) Q→2 calls Q invertible, but its rows are proportional ($\det = 0$); R→3 calls R singular, but $\det R = 1$.

Key takeaway. HHL needs a Hermitian (or embedded) and invertible matrix; check $A = A^\dagger$ and $\det A \neq 0$ first.

Q.4 SINGLE CORRECT · Condition number of a diagonal matrix

The condition number of $A = \mathrm{diag}(1,\ 0.5,\ 0.01,\ 0.2)$ is

(A) 0.01 **(B)** 1.71 **(C)** 50 **(D)** ✓ 100

Answer: (D) Detailed solution

Correct answer. (D) 100

Concepts required. For a diagonal matrix the eigenvalues and singular values are the absolute values of the diagonal entries.

Relevant formulae. $\kappa = \|A\|_2\|A^{-1}\|_2 = \sigma_{\max}/\sigma_{\min}$; diagonal: $\sigma_i = |a_{ii}|$.

Step-by-step explanation. For a diagonal (hence Hermitian) matrix the eigenvalues are the diagonal entries: $1, 0.5, 0.01, 0.2$.

$$\kappa = \frac{|\lambda|_{\max}}{|\lambda|_{\min}} = \frac{1}{0.01} = 100.$$

Equivalently $\kappa = \|A\|_2\,\|A^{-1}\|_2 = 1 \times 100$. A system with $\kappa = 100$ amplifies relative errors in $\vec{b}$ by up to a factor 100; for HHL the run time grows with κ (Section D).

Why the other options are incorrect.

(A) 0.01 is $\lambda_{\min}/\lambda_{\max}$, the reciprocal; $\kappa \geq 1$ always.

(B) 1.71 is the sum of the entries (the trace), which has nothing to do with conditioning.

(C) 50 divides 0.5 by 0.01, using the second-largest instead of the largest entry.

Key takeaway. κ = largest over smallest $|\lambda|$; $\kappa \geq 1$, and $\kappa = 1$ only for multiples of unitaries.

Q.5 SINGLE CORRECT · Eigenvalues of a complex Hermitian matrix

The Hermitian matrix $A = \begin{pmatrix} 3 & 1-i \\ 1+i & 2 \end{pmatrix}$ has eigenvalues

(A) 3 and 2 **(B)** 0 and 5 **(C)** ✓ 1 and 4 **(D)** $2 \pm i$

Answer: (C) Detailed solution

Correct answer. (C) 1 and 4

Concepts required. Hermitian matrices may have complex entries but always have real eigenvalues; the off-diagonal pair contributes $-|a_{12}|^2$ to the determinant.

Relevant formulae. $\det\left(\begin{smallmatrix} a & z \\ \bar{z} & d \end{smallmatrix}\right) = ad - |z|^2$; $\lambda_{1,2} = \frac{\mathrm{tr}\,A \pm \sqrt{(\mathrm{tr}\,A)^2 - 4\det A}}{2}$.

Step-by-step explanation. $\mathrm{tr}\,A = 5$, $\det A = 3\cdot 2 - (1-i)(1+i) = 6 - (1+1) = 4$. The eigenvalues solve $\lambda^2 - 5\lambda + 4 = 0$: $\lambda = 1$ or $\lambda = 4$. Both are real, as for every Hermitian matrix, even though A has complex entries. (Note $(1-i)(1+i) = |1+i|^2 = 2$.) A is positive definite with $\kappa = 4$.

Why the other options are incorrect.

(A) 3 and 2 are the diagonal entries; they ignore the coupling $1 \mp i$ (product $6 \neq \det A = 4$).

(B) 0 and 5 would need $\det A = 0$, i.e. $|1+i|^2 = 6$; in fact $|1+i|^2 = 2$.

(D) $2 \pm i$ are complex, which is impossible for a Hermitian matrix (their sum 4 is not even $\mathrm{tr}\,A = 5$); the discriminant $25 - 16 = 9$ is positive.

Key takeaway. For a Hermitian 2×2 matrix: $\det = ad - |z|^2$, and the eigenvalues must come out real.

Q.6 SINGLE CORRECT · Inverse via spectral decomposition

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenpairs $\lambda_1 = 3$, $|u_1\rangle = \frac{1}{\sqrt{2}}(1,1)$ and $\lambda_2 = 1$, $|u_2\rangle = \frac{1}{\sqrt{2}}(1,-1)$. Using $A^{-1} = \sum_j \lambda_j^{-1} |u_j\rangle\langle u_j|$, the off-diagonal entry $(A^{-1})_{12}$ is

(A) ✓ $-\frac{1}{3}$ **(B)** $+\frac{1}{3}$ **(C)** $-\frac{2}{3}$ **(D)** $\frac{1}{6}$

Answer: (A) Detailed solution

Correct answer. **(A)** $-\frac{1}{3}$

Concepts required. Spectral decomposition: a Hermitian matrix is a weighted sum of projectors onto its eigenvectors; functions of A act on the weights.

Relevant formulae. $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$, $f(A) = \sum_j f(\lambda_j) |u_j\rangle\langle u_j|$, so $A^{-1} = \sum_j \lambda_j^{-1} |u_j\rangle\langle u_j|$.

Step-by-step explanation. $|u_1\rangle\langle u_1| = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $|u_2\rangle\langle u_2| = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. Hence

$$A^{-1} = \frac{1}{3}\cdot\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{1}\cdot\frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{6}+\frac{1}{2} & \frac{1}{6}-\frac{1}{2} \\ \frac{1}{6}-\frac{1}{2} & \frac{1}{6}+\frac{1}{2} \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

So $(A^{-1})_{12} = -\frac{1}{3}$, in agreement with the 2×2 formula $\frac{1}{\det A}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, $\det A = 3$. This "invert every eigenvalue" formula is exactly what HHL implements on a quantum state.

Why the other options are incorrect.

(B) $+\frac{1}{3}$ has the wrong sign: the $\lambda = 1$ projector contributes $-\frac{1}{2}$ off the diagonal and dominates $+\frac{1}{6}$.

(C) $-\frac{2}{3}$ forgets the factor $\frac{1}{2}$ in $|u_j\rangle\langle u_j|$ for one of the terms $(\frac{1}{3} - 1)$.

(D) $\frac{1}{6}$ keeps only the $\lambda = 3$ term and forgets the $\lambda = 1$ projector.

Key takeaway. $f(A)$ = same eigenvectors, f applied to the eigenvalues. HHL applies $f(\lambda) = 1/\lambda$ to a quantum state.

Q.7 ASSERTION–REASON · Spectral theorem vs. real eigenvalues

Assertion (A): Every Hermitian matrix can be diagonalised by a unitary matrix, $A = V\Lambda V^\dagger$.

Reason (R): All eigenvalues of a Hermitian matrix are real.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) ✓ Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (B) Detailed solution

Correct answer. **(B)** Both A and R are true, but R is *not* the correct explanation of A

Concepts required. Spectral theorem: normal matrices ($AA^\dagger = A^\dagger A$) are exactly the unitarily diagonalisable ones. Hermitian matrices are normal and have real eigenvalues.

Relevant formulae. $A = V\Lambda V^\dagger$, $V^\dagger V = I$; $\lambda\langle u|u\rangle = \langle u|A|u\rangle = \bar{\lambda}\langle u|u\rangle$ for Hermitian A.

Step-by-step explanation. **A is true:** this is the spectral theorem (it holds for every *normal* matrix, $AA^\dagger = A^\dagger A$, and Hermitian matrices are normal). **R is true:** $\lambda\langle u|u\rangle = \langle u|A|u\rangle = \langle u|A^\dagger|u\rangle^* = \bar{\lambda}\langle u|u\rangle$, so $\lambda = \bar{\lambda}$. But R does not explain A: real eigenvalues do not imply diagonalisability — $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has the real eigenvalue 1 twice and cannot be diagonalised, and unitary matrices are diagonalisable by a unitary although their eigenvalues are complex. The explanation of A is normality. Answer: (B).

Why the other options are incorrect.

(A) claims R explains A, but real eigenvalues do not imply diagonalisability: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is a counter-

example.

(C) claims R is false, but every Hermitian matrix has real eigenvalues.

(D) claims A is false, but the spectral theorem guarantees a unitary diagonalisation for Hermitian matrices.

Key takeaway. Two separate facts about Hermitian matrices: unitary diagonalisability (from normality) and real eigenvalues (from $A = A^\dagger$). HHL needs both.

Q.8 STATEMENT I/II · Condition number of U and of A^{-1}

Statement I: Every unitary matrix has condition number $\kappa = 1$.

Statement II: For every invertible matrix A, $\kappa(A^{-1}) = \kappa(A)$.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true

Concepts required. The condition number is the ratio of the largest to the smallest singular value.

Relevant formulae. $\kappa(A) = \sigma_{\max}/\sigma_{\min}$; singular values of U unitary $= 1$; singular values of A^{-1} $= 1/\sigma_i$.

Step-by-step explanation. $\kappa(A) = \|A\|_2\|A^{-1}\|_2 = \sigma_{\max}/\sigma_{\min}$. **I is true:** a unitary U preserves lengths, so all its singular values equal 1 and $\kappa = 1/1 = 1$. **II is true:** the singular values of A^{-1} are $1/\sigma_i$, so $\kappa(A^{-1}) = \frac{1/\sigma_{\min}}{1/\sigma_{\max}} = \kappa(A)$ (directly: $\|A^{-1}\|\,\|(A^{-1})^{-1}\| = \|A^{-1}\|\,\|A\|$). Answer: (A).

Why the other options are incorrect.

(B) "both false" fails because both statements follow directly from the singular values.

(C) rejects II, but inverting swaps the largest and smallest singular values and leaves the ratio unchanged.

(D) rejects I, but a unitary preserves all lengths: $\sigma_{\max} = \sigma_{\min} = 1$.

Key takeaway. Unitaries are perfectly conditioned; inverting a matrix does not change its conditioning.

Q.9 SINGLE CORRECT · Sparsity of the 1-D Laplacian

The 8×8 matrix of the 1-D discrete Laplacian has 2 on the diagonal, -1 on the first upper and lower off-diagonals and 0 elsewhere. Its sparsity s (maximum non-zeros per row) and its total number of non-zero entries are

(A) $s = 8$; 64 **(B)** ✓ $s = 3$; 22 **(C)** $s = 2$; 22 **(D)** $s = 3$; 24

Answer: (B) Detailed solution

Correct answer. **(B)** $s = 3$; 22

Concepts required. Sparsity s counts the maximum number of non-zero entries per row; banded matrices have constant s.

Relevant formulae. Tridiagonal $n \times n$: $\text{nnz} = n + 2(n-1) = 3n - 2$; $s = 3$ for $n \geq 3$.

Step-by-step explanation. Interior rows $2, \ldots, 7$ contain $-1, 2, -1$: three non-zeros. The first and last rows contain only two ($2, -1$ and $-1, 2$). So $s = 3$. Total: diagonal 8 + upper off-diagonal 7 + lower off-diagonal 7 = 22 (not $3 \times 8 = 24$, because the boundary rows have only two entries). Such matrices are *sparse*: s stays 3 for any size d, which is the setting in which HHL's cost $\propto s$ (or

s^2) is small.

Why the other options are incorrect.

(A) $s = 8$, 64 treats the matrix as dense; most entries are 0.

(C) $s = 2$ counts only the off-diagonals and forgets the diagonal 2.

(D) $24 = 3 \times 8$ assumes every row has three non-zeros, but the first and last rows have only two.

Key takeaway. Sparse means $s \ll d$ non-zeros per row; HHL's cost depends on s, not on d.

Q.10 SINGLE CORRECT · Spectrum and κ of the Laplacian

The eigenvalues of the $n \times n$ Laplacian of Q.9 (with 2 on the diagonal) are $\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$, $k = 1, \ldots, n$. For $n = 3$ the condition number is

(A) ✓ $3 + 2\sqrt{2} \approx 5.83$ **(B)** 2 **(C)** $\sqrt{2} \approx 1.41$ **(D)** $1 + \sqrt{2} \approx 2.41$

Answer: (A) Detailed solution

Correct answer. (A) $3 + 2\sqrt{2} \approx 5.83$

Concepts required. The discrete Laplacian has known eigenvalues; its smallest eigenvalue shrinks like $1/n^2$ as the grid is refined.

Relevant formulae. $\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$; $\kappa = \frac{\lambda_n}{\lambda_1}$; $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $(2+\sqrt{2})(2-\sqrt{2}) = 2$.

Step-by-step explanation. For $n = 3$: $\frac{k\pi}{4}$ with $k = 1, 2, 3$ gives $\lambda_1 = 2 - 2 \cdot \frac{\sqrt{2}}{2} = 2 - \sqrt{2} \approx 0.586$, $\lambda_2 = 2 - 0 = 2$, $\lambda_3 = 2 + \sqrt{2} \approx 3.414$. Hence

$$\kappa = \frac{2+\sqrt{2}}{2-\sqrt{2}} = \frac{(2+\sqrt{2})^2}{4-2} = \frac{6+4\sqrt{2}}{2} = 3 + 2\sqrt{2} \approx 5.83.$$

For large n, $\lambda_1 \approx (\pi/(n+1))^2$ and $\kappa \approx 4(n+1)^2/\pi^2$: discretised differential operators become badly conditioned as the grid is refined, which limits HHL's advantage for such problems.

Why the other options are incorrect.

(B) 2 is the middle eigenvalue λ_2, not a ratio.

(C) $\sqrt{2}$ is λ_2/λ_1 evaluated wrongly; the ratio of the extreme eigenvalues is needed.

(D) $1 + \sqrt{2}$ is $\sqrt{\kappa}$; it misses the square that appears when rationalising $\frac{2+\sqrt{2}}{2-\sqrt{2}}$.

Key takeaway. Finer discretisations of differential equations have $\kappa \propto n^2$: the κ factor of HHL can eat the speed-up.

Q.11 SINGLE CORRECT · Normalisation and amplitude encoding

The vector $\vec{b} = (3, 0, 4, 0)$ is amplitude-encoded in two qubits (index = binary number $q_0 q_1$). Measuring both qubits, the probability of the outcome $|10\rangle$ is

(A) 0.16 **(B)** ✓ 0.64 **(C)** 0.36 **(D)** 0.8

Answer: (B) Detailed solution

Correct answer. (B) 0.64

Concepts required. Amplitude encoding stores the normalised vector $\vec{b}/\|\vec{b}\|$ as amplitudes; probabilities are squared magnitudes.

Relevant formulae. $|b\rangle = \sum_i \frac{b_i}{\|\vec{b}\|}|i\rangle$; $P(i) = \frac{|b_i|^2}{\|\vec{b}\|^2}$; index i in binary $q_0 q_1$.

Step-by-step explanation. Normalise: $\|\vec{b}\| = \sqrt{9+16} = 5$, so $|b\rangle = \frac{3}{5}|00\rangle + \frac{4}{5}|10\rangle$ (index $2 = 10_2$ has amplitude $4/5$). Born rule: $P(10) = |4/5|^2 = 16/25 = 0.64$. The norm $\|\vec{b}\| = 5$ is *not* stored in the state; it must be kept classically (the notebook multiplies it back at the end).

Why the other options are incorrect.

(A) $0.16 = 0.4^2$ divides the entry 4 by 10 instead of by the norm 5 before squaring.

(C) 0.36 is the probability of $|00\rangle$ (amplitude 3/5), not of $|10\rangle$.

(D) 0.8 is the amplitude 4/5, not the probability; the Born rule squares it.

Key takeaway. Normalise first, then square: $P(i) = |b_i|^2/\|\vec{b}\|^2$. The norm itself is not in the state.

Q.12 SINGLE CORRECT · Solving $A\vec{x} = \vec{b}$ in the eigenbasis

With the eigenpairs of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ given in Q.6 and $\vec{b} = (1, 0)$, write $\vec{b} = \sum_j \beta_j |u_j\rangle$. Then $\vec{x} = \sum_j (\beta_j/\lambda_j)\,|u_j\rangle$ equals

(A) $(2,\ -1)$ **(B)** $\left(\frac{1}{2},\ -\frac{1}{2}\right)$ **(C)** ✓ $\left(\frac{2}{3},\ -\frac{1}{3}\right)$ **(D)** $\left(\frac{2}{3},\ \frac{1}{3}\right)$

Answer: (C) **Detailed solution**

Correct answer. (C) $\left(\frac{2}{3},\ -\frac{1}{3}\right)$

Concepts required. Solving through the eigenbasis: expand $\vec{b}$, divide each coefficient by its eigenvalue, recombine.

Relevant formulae. $\beta_j = \langle u_j|b\rangle$; $\vec{x} = A^{-1}\vec{b} = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle$.

Step-by-step explanation. $\beta_1 = \langle u_1|b\rangle = \frac{1}{\sqrt{2}}$, $\beta_2 = \langle u_2|b\rangle = \frac{1}{\sqrt{2}}$. Divide by the eigenvalues 3 and 1:

$$\vec{x} = \frac{1}{3\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} + \frac{1}{2} \\ \frac{1}{6} - \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

Check: $A\vec{x} = (\frac{4}{3} - \frac{1}{3},\ \frac{2}{3} - \frac{2}{3}) = (1, 0)$. HHL performs these three steps (expand in the eigenbasis, divide by λ_j, recombine) coherently: phase estimation, controlled rotation, uncomputation.

Why the other options are incorrect.

(A) $(2, -1)$ multiplies by λ instead of dividing, or forgets the normalisation of the eigenvectors; $A(2, -1) = (3, 0)$.

(B) $(\frac{1}{2}, -\frac{1}{2})$ keeps only the $\lambda = 1$ component and drops the $\lambda = 3$ one.

(D) $(\frac{2}{3}, \frac{1}{3})$ has the wrong sign in the second component: $A(\frac{2}{3}, \frac{1}{3}) = (\frac{5}{3}, \frac{4}{3}) \neq (1, 0)$.

Key takeaway. $A^{-1}\vec{b} = \sum_j (\beta_j/\lambda_j)\,|u_j\rangle$ is the formula HHL realises with phase estimation and a controlled rotation.

Q.13 MATCHING LISTS · Norms, κ and determinant

For $A = \mathrm{diag}(4,\ 1,\ 0.5,\ 3)$ match each quantity (List-I) with its value (List-II).

List-I		List-II	
(P)	spectral norm $\|A\|_2$	**(1)**	8
(Q)	spectral norm $\|A^{-1}\|_2$	**(2)**	4
(R)	condition number $\kappa(A)$	**(3)**	6
(S)	determinant $\det A$	**(4)**	2

(A) P→2, Q→4, R→3, S→1 **(B)** P→2, Q→3, R→1, S→4

(C) P→1, Q→4, R→2, S→3 **(D)** ✓ P→2, Q→4, R→1, S→3

Answer: (D) **Detailed solution**

Correct answer. **(D)** P→2, Q→4, R→1, S→3
Concepts required. For a diagonal matrix the spectral norm is the largest |entry|; norms of A^{-1} use reciprocals; det is the product.
Relevant formulae. $\|A\|_2 = \max|\lambda|$, $\|A^{-1}\|_2 = 1/\min|\lambda|$, $\kappa = \|A\|_2\|A^{-1}\|_2$, $\det A = \prod \lambda_i$.
Step-by-step explanation.

- P: $\|A\|_2$ = largest $|\lambda| = 4$ (2).
- Q: $A^{-1} = \mathrm{diag}(\frac{1}{4}, 1, 2, \frac{1}{3})$, so $\|A^{-1}\|_2 = 1/|\lambda|_{\min} = 1/0.5 = 2$ (4).
- R: $\kappa = \|A\|_2\|A^{-1}\|_2 = 4 \times 2 = 8$ (1).
- S: $\det A = 4 \cdot 1 \cdot 0.5 \cdot 3 = 6$ (3).

Note that the determinant says nothing about conditioning: $\det(0.1\, I_{10}) = 10^{-10}$ is tiny while $\kappa = 1$.
Why the other options are incorrect.
(A) R→3 gives $\kappa = 6$ (the determinant) and S→1 gives $\det A = 8$.
(B) Q→3 gives $\|A^{-1}\|_2 = 6$ and S→4 gives $\det A = 2$: the determinant and the inverse norm are exchanged.
(C) P→1 gives $\|A\|_2 = 8$ (that is κ) and R→2 gives $\kappa = 4$ (that is $\|A\|_2$).
Key takeaway. $\kappa = \|A\|\,\|A^{-1}\|$; the determinant is unrelated to the condition number.

Q.14 SINGLE CORRECT · Hermitian embedding of a non-Hermitian matrix

HHL needs a Hermitian matrix, so the non-Hermitian $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ is embedded as $\tilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}$ (4×4). The eigenvalues of $\tilde{A}$ are

(A) $3 \pm 2\sqrt{2}$
(B) ± 1, each twice
(C) $1, 1, 1, 1$
(D) ✓ $\pm(\sqrt{2}+1)$ and $\pm(\sqrt{2}-1)$

Answer: (D) **Detailed solution**

Correct answer. **(D)** $\pm(\sqrt{2}+1)$ and $\pm(\sqrt{2}-1)$
Concepts required. The Hermitian embedding $\tilde{A} = \left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm\sigma_i(A)$.
Relevant formulae. $\tilde{A}^2 = \mathrm{diag}(AA^\dagger, A^\dagger A)$; $\sigma_i = \sqrt{\mathrm{eig}(A^\dagger A)}$; $3 \pm 2\sqrt{2} = (\sqrt{2} \pm 1)^2$.
Step-by-step explanation. $\tilde{A}^2 = \left(\begin{smallmatrix} AA^\dagger & 0 \\ 0 & A^\dagger A \end{smallmatrix}\right)$, so the eigenvalues of $\tilde{A}$ are $\pm\sigma_i$, the singular values of A with both signs. $A^\dagger A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 5 \end{smallmatrix}\right)$ has trace 6 and determinant 1: eigenvalues $3 \pm 2\sqrt{2} = (\sqrt{2} \pm 1)^2$. Hence $\sigma = \sqrt{2} \pm 1$ and the spectrum of $\tilde{A}$ is $\{\pm(\sqrt{2}+1), \pm(\sqrt{2}-1)\} \approx \{\pm 2.414, \pm 0.414\}$. The eigenvalues of A itself $(1, 1)$ are irrelevant here; $\kappa(\tilde{A}) = \frac{\sqrt{2}+1}{\sqrt{2}-1} = 3 + 2\sqrt{2} \approx 5.83$, and HHL must handle *negative* eigenvalues.
Why the other options are incorrect.
(A) $3 \pm 2\sqrt{2}$ are the eigenvalues of $A^\dagger A$, i.e. the *squares* of the singular values (and they are never negative).
(B) ± 1 twice again uses the eigenvalues of A; the embedding is governed by singular values.
(C) $1, 1, 1, 1$ uses the eigenvalues of A (a defective matrix), not its singular values, and forgets the $\pm$ pairs.
Key takeaway. Embedding a non-Hermitian A: dimension doubles, eigenvalues become $\pm\sigma_i$, and $\kappa = \sigma_{\max}/\sigma_{\min}$.

Q.15 STATEMENT I/II · When is e^{iAt} unitary?

Statement I: If A is Hermitian, then $U = e^{iAt}$ is unitary for every real t.

Statement II: If A is any real square matrix, then e^{iAt} is unitary for every real t.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. (C) Statement I is true, but Statement II is false

Concepts required. e^{iAt} is unitary iff $(e^{iAt})^\dagger e^{iAt} = I$; this holds for every t exactly when A is Hermitian.

Relevant formulae. $(e^{iAt})^\dagger = e^{-iA^\dagger t}$; $A = V\Lambda V^\dagger \Rightarrow e^{iAt} = Ve^{i\Lambda t}V^\dagger$.

Step-by-step explanation. **I is true:** with $A = V\Lambda V^\dagger$, $e^{iAt} = V\,\mathrm{diag}(e^{i\lambda_j t})V^\dagger$ and $|e^{i\lambda_j t}| = 1$ because λ_j is real; equivalently $(e^{iAt})^\dagger = e^{-iA^\dagger t} = e^{-iAt}$. **II is false:** realness is not enough. For $A = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, $e^{iAt} = I + iAt = \left(\begin{smallmatrix}1&it\\0&1\end{smallmatrix}\right)$, which maps $(0,1)$ to $(it,1)$ of length $\sqrt{1+t^2} \neq 1$. Only real *symmetric* matrices are guaranteed to work. This is why HHL embeds non-Hermitian matrices first. Answer: (C).

Why the other options are incorrect.

(A) accepts II, but a real non-symmetric matrix such as $\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$ gives a non-unitary e^{iAt}.

(B) rejects I, but Hermitian A has real eigenvalues, so $e^{i\lambda t}$ are pure phases.

(D) rejects I and accepts II — both judgements are reversed.

Key takeaway. Hermitian $\Leftrightarrow$ e^{iAt} unitary. This is the reason HHL requires (or embeds into) a Hermitian matrix.

Q.16 SINGLE CORRECT · Error amplification by κ

A linear system has $\kappa(A) = 1000$. The right-hand side is known with relative error $\|\delta\vec{b}\|/\|\vec{b}\| = 10^{-6}$. The best general upper bound on the relative error $\|\delta\vec{x}\|/\|\vec{x}\|$ of the solution is

(A) ✓ 10^{-3} **(B)** 10^{-6}

(C) 10^{-9} **(D)** $10^{-6}/\sqrt{1000} \approx 3 \times 10^{-8}$

Answer: (A) Detailed solution

Correct answer. (A) 10^{-3}

Concepts required. Perturbation theory for linear systems: the relative error in $\vec{x}$ can be κ times the relative error in $\vec{b}$.

Relevant formulae. $\dfrac{\|\delta\vec{x}\|}{\|\vec{x}\|} \le \kappa(A)\,\dfrac{\|\delta\vec{b}\|}{\|\vec{b}\|}$.

Step-by-step explanation. From $A(\vec{x}+\delta\vec{x}) = \vec{b}+\delta\vec{b}$: $\delta\vec{x} = A^{-1}\delta\vec{b}$, so $\|\delta\vec{x}\| \le \|A^{-1}\|\,\|\delta\vec{b}\|$, and $\|\vec{b}\| \le \|A\|\,\|\vec{x}\|$. Dividing:

$$\frac{\|\delta\vec{x}\|}{\|\vec{x}\|} \le \|A\|\|A^{-1}\|\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|} = \kappa\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|} = 1000 \times 10^{-6} = 10^{-3}.$$

The bound is attained when $\vec{b}$ points along the largest and $\delta\vec{b}$ along the smallest singular direction. κ measures the intrinsic sensitivity of the problem — no algorithm, classical or quantum, removes it.

Why the other options are incorrect.

(B) 10^{-6} assumes errors are not amplified, true only for $\kappa = 1$.

(C) 10^{-9} divides by κ instead of multiplying.

(D) $10^{-6}/\sqrt{1000}$ has no basis: κ multiplies the error bound.
Key takeaway. κ is an intrinsic sensitivity: $\log_{10}\kappa$ digits can be lost, whatever the algorithm.

Paragraph for Questions 17 and 18

The notebook `HHL.ipynb` lists the 2×2 example

$$A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The questions below concern the classical linear algebra that HHL relies on for this system.

Q.17 Paragraph based · 2×2 example: eigenvalues

The eigenvalues of A and its condition number are

(A) $\frac{1}{3}$ and 1; $\kappa = 3$
(B) 1 and $-\frac{1}{3}$; $\kappa = 3$
(C) $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = \frac{4}{3}$
(D) ✓ $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$

Answer: (D) Detailed solution

Correct answer. **(D)** $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$
Concepts required. For $\left(\begin{smallmatrix} a & c \\ c & a \end{smallmatrix}\right)$ the eigenvalues are $a \pm c$ with eigenvectors $\frac{1}{\sqrt{2}}(1, \pm 1)$.
Relevant formulae. $\lambda = a \pm c$; $\kappa = \frac{|\lambda|_{\max}}{|\lambda|_{\min}}$; here $a = 1$, $c = -\frac{1}{3}$.
Step-by-step explanation. $\operatorname{tr} A = 2$, $\det A = 1 - \frac{1}{9} = \frac{8}{9}$. The eigenvalues solve $\lambda^2 - 2\lambda + \frac{8}{9} = 0$: $\lambda = 1 \pm \sqrt{1 - \frac{8}{9}} = 1 \pm \frac{1}{3}$, i.e. $\frac{2}{3}$ (eigenvector $\frac{1}{\sqrt{2}}(1, 1)$) and $\frac{4}{3}$ (eigenvector $\frac{1}{\sqrt{2}}(1, -1)$).

$$\kappa = \frac{4/3}{2/3} = 2.$$

(The diagonal 1 and the off-diagonal $-\frac{1}{3}$ are entries, not eigenvalues.) With $\kappa = 2$ the notebook's "auto" time maps $\frac{2}{3} \to$ clock value 1 and $\frac{4}{3} \to$ clock value 2, so HHL is exact for this matrix.
Why the other options are incorrect.
(A) $\frac{1}{3}$ and 1 give the trace $\frac{4}{3} \neq 2$; they are not roots of the characteristic polynomial.
(B) 1 and $-\frac{1}{3}$ are the matrix entries, not eigenvalues (their product $-\frac{1}{3} \neq \det A = \frac{8}{9}$).
(C) $\kappa = \frac{4}{3}$ divides the largest eigenvalue by 1 (the diagonal entry) instead of by $\lambda_{\min} = \frac{2}{3}$.
Key takeaway. Symmetric 2×2 with equal diagonal: eigenvalues $a \pm c$ — a two-second calculation worth memorising.

Q.18 Paragraph based · 2×2 example: normalised solution

The solution $\vec{x} = A^{-1}\vec{b}$ and the probability of measuring the system qubit as $|0\rangle$ in the normalised state $|x\rangle$ are

(A) $\vec{x} = \left(1,\ \frac{1}{3}\right)$; $\frac{1}{\sqrt{10}}$
(B) $\vec{x} = \left(\frac{9}{8},\ \frac{3}{8}\right)$; 0.75
(C) ✓ $\vec{x} = \left(\frac{9}{8},\ \frac{3}{8}\right)$; 0.9
(D) $\vec{x} = \left(\frac{9}{8},\ -\frac{3}{8}\right)$; 0.9

Answer: (C) Detailed solution

Correct answer. **(C)** $\vec{x} = \left(\frac{9}{8},\ \frac{3}{8}\right)$; 0.9
Concepts required. The normalised quantum state $|x\rangle$ keeps the ratios of the solution components; probabilities are squared normalised components.

Relevant formulae. $\vec{x} = A^{-1}\vec{b}$; $|x\rangle = \vec{x}/\|\vec{x}\|$; $P(i) = x_i^2/\|\vec{x}\|^2$.

Step-by-step explanation. $A^{-1} = \frac{1}{\det A}\begin{pmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & 1 \end{pmatrix} = \frac{9}{8}\begin{pmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & 1 \end{pmatrix}$, so $\vec{x} = A^{-1}(1,0)^T = \left(\frac{9}{8}, \frac{3}{8}\right)$. Check: $\frac{9}{8} - \frac{1}{3}\cdot\frac{3}{8} = \frac{9}{8} - \frac{1}{8} = 1$ and $-\frac{1}{3}\cdot\frac{9}{8} + \frac{3}{8} = 0$. Normalised: $\vec{x} \propto (3,1)$, $|x\rangle = \frac{1}{\sqrt{10}}(3\,|0\rangle + |1\rangle)$, so $P(0) = \frac{9}{10} = 0.9$. The *ratio* $x_1/x_2 = 3$ survives in the quantum state; the overall scale $\|\vec{x}\| = \frac{3}{8}\sqrt{10}$ does not.

Why the other options are incorrect.

(A) $(1, \frac{1}{3})$ is $\vec{x}$ up to the factor $\frac{9}{8}$, and $\frac{1}{\sqrt{10}}$ is an amplitude, not the probability of $|0\rangle$.

(B) 0.75 is $x_0/(x_0 + x_1) = \frac{9/8}{12/8}$: it normalises the components linearly instead of by squares.

(D) $(\frac{9}{8}, -\frac{3}{8})$ has the wrong sign: the off-diagonal $-\frac{1}{3}$ of A becomes $+\frac{1}{3}$ in A^{-1}.

Key takeaway. Measuring $|x\rangle$ gives $x_i^2/\|\vec{x}\|^2$: ratios survive, the scale does not.

Q.19 SINGLE CORRECT · Embedding + padding: qubit count

A 5×5 *non-Hermitian* matrix is prepared for HHL exactly as in the notebook: first the Hermitian embedding $\begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}$, then padding with an identity block to the next power of 2. The number of system qubits n_b is

(A) 5 **(B)** ✓ 4 **(C)** 10 **(D)** 3

Answer: (B) Detailed solution

Correct answer. **(B)** 4

Concepts required. Two preparation steps change the size: Hermitian embedding ($d \to 2d$) and padding to a power of two.

Relevant formulae. $n_b = \lceil \log_2(2d) \rceil$ for non-Hermitian A; $n_b = \lceil \log_2 d \rceil$ for Hermitian A.

Step-by-step explanation. Embedding doubles the dimension: $2 \times 5 = 10$. The next power of two is $16 = 2^4$; the extra 6 dimensions are filled with an identity block (eigenvalue 1, decoupled from the system because $\vec{b}$ is padded with zeros). Hence $n_b = \log_2 16 = 4$ system qubits. (Without the embedding $5 \to 8$ would need 3 qubits.) In general $n_b = \lceil \log_2(2d) \rceil$ for a non-Hermitian $d \times d$ matrix.

Why the other options are incorrect.

(A) 5 counts one qubit per row; qubit counts are logarithmic in the dimension.

(C) 10 is the embedded dimension, not the number of qubits.

(D) 3 pads $5 \to 8$ but forgets the doubling of the embedding.

Key takeaway. Qubits = $\log_2$(dimension after embedding and padding).

Q.20 ASSERTION–REASON · κ from singular values

Assertion (A): For a Hermitian positive-definite matrix the condition number equals $\lambda_{\max}/\lambda_{\min}$.

Reason (R): The singular values of a Hermitian matrix are the absolute values of its eigenvalues.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. Singular values are the square roots of the eigenvalues of $A^\dagger A$; for Hermitian A they are $|\lambda_j|$.
Relevant formulae. $\kappa = \sigma_{\max}/\sigma_{\min}$; $A = V\Lambda V^\dagger \Rightarrow \sigma_j = |\lambda_j|$; positive definite: $\lambda_j > 0$.
Step-by-step explanation. By definition $\kappa = \sigma_{\max}/\sigma_{\min}$ (spectral norm of A times that of A^{-1}). **R is true:** for $A = V\Lambda V^\dagger$, $A^\dagger A = V\Lambda^2 V^\dagger$, so $\sigma_j = \sqrt{\lambda_j^2} = |\lambda_j|$. **A is true**, and it follows from R: for a positive-definite matrix all $\lambda_j > 0$, so $|\lambda_j| = \lambda_j$ and $\kappa = \sigma_{\max}/\sigma_{\min} = \lambda_{\max}/\lambda_{\min}$. R is the correct explanation. Answer: (A).
Why the other options are incorrect.
(B) denies the explanatory link, but A is derived directly from R plus positivity.
(C) rejects R, but $A^\dagger A = A^2 = V\Lambda^2 V^\dagger$ gives $\sigma_j = |\lambda_j|$.
(D) rejects A, but for positive eigenvalues $|\lambda| = \lambda$ and the ratio is $\lambda_{\max}/\lambda_{\min}$.
Key takeaway. For Hermitian matrices $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$, the definition used by the notebook.

SECTION B · Quantum Computing Foundations Q.21–40

Q.21 SINGLE CORRECT · Born rule

A qubit is in the state $|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2i\,|1\rangle)$. The probability of measuring 1 is

(A) $\frac{2}{\sqrt{5}}$ **(B)** $\frac{2}{5}$ **(C)** ✓ $\frac{4}{5}$ **(D)** $-\frac{4}{5}$

Answer: (C) **Detailed solution**

Correct answer. **(C)** $\frac{4}{5}$

Concepts required. Born rule: the probability of an outcome is the squared magnitude of its amplitude.

Relevant formulae. $P(k) = |\langle k|\psi\rangle|^2$; $|a+ib|^2 = a^2+b^2$; $\sum_k P(k) = 1$.

Step-by-step explanation. Born rule: $P(1) = |\langle 1|\psi\rangle|^2 = \left|\frac{2i}{\sqrt{5}}\right|^2 = \frac{|2i|^2}{5} = \frac{4}{5}$. Check the normalisation: $\frac{1}{5}(|1|^2 + |2i|^2) = \frac{1+4}{5} = 1$. The factor i is a *relative phase*: it does not change the computational-basis probabilities but would change the outcome of a measurement in the X basis.

Why the other options are incorrect.

(A) $\frac{2}{\sqrt{5}}$ is the magnitude of the amplitude, not the probability.

(B) $\frac{2}{5}$ uses $|2i|$ instead of $|2i|^2$ in the numerator.

(D) $-\frac{4}{5}$ squares $2i$ as $(2i)^2 = -4$ instead of taking $|2i|^2 = 4$; probabilities are never negative.

Key takeaway. Probability = |amplitude|^2; phases such as i drop out of computational-basis statistics.

Q.22 SINGLE CORRECT · State-vector size and memory

The default HHL circuit uses 7 qubits. A state-vector simulator stores one complex double-precision number (16 bytes) per amplitude. The number of amplitudes and the memory of the state are

(A) ✓ 128 amplitudes; 2048 bytes **(B)** 7 amplitudes; 112 bytes

(C) 128 amplitudes; 1024 bytes **(D)** 49 amplitudes; 784 bytes

Answer: (A) **Detailed solution**

Correct answer. **(A)** 128 amplitudes; 2048 bytes

Concepts required. An n-qubit state has 2^n complex amplitudes; a simulator stores all of them.

Relevant formulae. memory $= 2^n \times 16$ bytes (fp64 complex), $2^n \times 8$ bytes (fp32).

Step-by-step explanation. An n-qubit state has 2^n amplitudes: $2^7 = 128$. Memory $= 16 \times 2^7 = 2048$ bytes $= 2$ KiB. In general 16×2^n bytes: every extra qubit doubles the memory, 30 qubits need 17.2 GB and 40 qubits 17.6 TB. (A real `complex128` has 8 bytes for the real part and 8 for the imaginary part; 1024 bytes would be single precision.)

Why the other options are incorrect.

(B) 7 amplitudes treats qubits like classical bits with one number each.

(C) 1024 bytes assumes 8 bytes per amplitude (single precision), not 16.

(D) $49 = 7^2$ has no meaning here; the count is 2^7.

Key takeaway. Memory doubles with every qubit: this, not the gate count, limits classical simulation of HHL.

Q.23 SINGLE CORRECT · Gate identity $HZH = X$

The product HZH equals

(A) Y **(B)** Z **(C)** ✓ X **(D)** $-X$

Answer: (C) Detailed solution

Correct answer. (C) X

Concepts required. Conjugating by H exchanges the X and Z axes.

Relevant formulae. $HZH = X$, $HXH = Z$, $HYH = -Y$, $H^2 = I$.

Step-by-step explanation.

$$HZH = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}0 & 2\\ 2 & 0\end{pmatrix} = X.$$

The Hadamard exchanges the Z and X bases ($H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$), so it maps Z to X and vice versa ($HXH = Z$).

Why the other options are incorrect.

(A) Y is the image of neither X nor Z under H-conjugation ($HYH = -Y$).

(B) Z would require H to commute with Z, which it does not.

(D) $-X$ has the wrong sign: the explicit product gives $+X$.

Key takeaway. H switches between the Z basis and the X basis; this is the basis of phase kickback and QPE.

Q.24 MATCHING LISTS · R_Y rotations of $|0\rangle$

Match each rotation applied to $|0\rangle$ (List-I) with the resulting state (List-II).

List-I		List-II	
(P)	$R_Y(\pi)\,\|0\rangle$	**(1)**	$\|+\rangle$
(Q)	$R_Y(\pi/2)\,\|0\rangle$	**(2)**	$\frac{\sqrt{3}}{2}\|0\rangle + \frac{1}{2}\|1\rangle$
(R)	$R_Y(-\pi/2)\,\|0\rangle$	**(3)**	$\|1\rangle$
(S)	$R_Y(\pi/3)\,\|0\rangle$	**(4)**	$\|-\rangle$

(A) P→1, Q→3, R→4, S→2 **(B)** ✓ P→3, Q→1, R→4, S→2

(C) P→3, Q→4, R→1, S→2 **(D)** P→3, Q→1, R→2, S→4

Answer: (B) Detailed solution

Correct answer. (B) P→3, Q→1, R→4, S→2

Concepts required. $R_Y(\theta)$ rotates the Bloch vector in the x–z plane; on $|0\rangle$ it gives real amplitudes $\cos\frac{\theta}{2}$ and $\sin\frac{\theta}{2}$.

Relevant formulae. $R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$.

Step-by-step explanation. $R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$ (first column of the matrix in the conventions).

- P: $\theta = \pi$: $\cos\frac{\pi}{2}\,|0\rangle + \sin\frac{\pi}{2}\,|1\rangle = |1\rangle$ (3).
- Q: $\theta = \frac{\pi}{2}$: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$ (1).
- R: $\theta = -\frac{\pi}{2}$: $\cos(-\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$, $\sin(-\frac{\pi}{4}) = -\frac{1}{\sqrt{2}}$, giving $|-\rangle$ (4).
- S: $\theta = \frac{\pi}{3}$: $\cos\frac{\pi}{6}\,|0\rangle + \sin\frac{\pi}{6}\,|1\rangle = \frac{\sqrt{3}}{2}\,|0\rangle + \frac{1}{2}\,|1\rangle$ (2).

In HHL the angle $\theta = 2\arcsin(C/\lambda)$ is chosen so that the amplitude of $|1\rangle$ is exactly $\sin\frac{\theta}{2} = C/\lambda$.

Why the other options are incorrect.

(A) P→1, Q→3 exchanges $R_Y(\pi)$ and $R_Y(\pi/2)$: the half-angle means $\theta = \pi$ is a full flip.

(C) Q→4, R→1 exchanges $|+\rangle$ and $|-\rangle$: a positive angle gives a positive $|1\rangle$ amplitude.

(D) R→2, S→4 assigns $|-\rangle$ to $\theta = \pi/3$ and the $\frac{\sqrt{3}}{2}, \frac{1}{2}$ state to $\theta = -\pi/2$.

Key takeaway. The half-angle: $R_Y(\theta)$ puts amplitude $\sin\frac{\theta}{2}$ on $|1\rangle$, hence HHL uses $\theta = 2\arcsin(C/\lambda)$.

Q.25 SINGLE CORRECT · Entanglement test

Which of the following two-qubit states is *entangled*?

(A) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

(B) $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$

(C) ✓ $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

(D) $\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$

Answer: (C) Detailed solution

Correct answer. (C) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

Concepts required. A two-qubit pure state is a product state iff its 2×2 coefficient matrix has rank 1 (determinant 0).

Relevant formulae. $|\psi\rangle = \sum a_{ij}|ij\rangle$ is a product $\iff a_{00}a_{11} - a_{01}a_{10} = 0$.

Step-by-step explanation. Write $|\psi\rangle = \sum a_{ij}|ij\rangle$. It is a product state iff the coefficient matrix $\left(\begin{smallmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{smallmatrix}\right)$ has rank 1, i.e. $a_{00}a_{11} - a_{01}a_{10} = 0$.

- $(1,1,1,-1)/2$: $1 \cdot (-1) - 1 \cdot 1 = -2 \neq 0$: **entangled**.
- $(1,1,1,1)/2 = |+\rangle|+\rangle$; $(1,-1,1,-1)/2 = |+\rangle|-\rangle$; $(1,1,-1,-1)/2 = |-\rangle|+\rangle$: determinant 0, product states.

In HHL, phase estimation entangles the clock with the system ($\sum_j \beta_j |u_j\rangle|k_j\rangle$); uncomputation removes this entanglement again.

Why the other options are incorrect.

(A) $\frac{1}{2}(1,1,1,1) = |+\rangle|+\rangle$: determinant $1 - 1 = 0$, a product state.

(B) $\frac{1}{2}(1,-1,1,-1) = |+\rangle|-\rangle$: determinant $-1 + 1 = 0$, a product state.

(D) $\frac{1}{2}(1,1,-1,-1) = |-\rangle|+\rangle$: determinant $-1 + 1 = 0$, a product state.

Key takeaway. Entanglement test for two qubits: $a_{00}a_{11} \neq a_{01}a_{10}$.

Q.26 SINGLE CORRECT · Phase kickback

The control qubit is in $|+\rangle$ and the target in an eigenstate $|u\rangle$ of U with $U|u\rangle = e^{i\phi}|u\rangle$. After a controlled-$U$ the joint state is

(A) $|+\rangle \otimes e^{i\phi}|u\rangle$ (the phase stays on the target)

(B) $\frac{1}{\sqrt{2}}\left(|0\rangle|u\rangle + |1\rangle U^\dagger|u\rangle\right)$, entangled

(C) $\frac{1}{\sqrt{2}}\left(e^{i\phi}|0\rangle + |1\rangle\right) \otimes |u\rangle$

(D) ✓ $\frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\phi}|1\rangle\right) \otimes |u\rangle$

Answer: (D) Detailed solution

Correct answer. (D) $\frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\phi}|1\rangle\right) \otimes |u\rangle$

Concepts required. Phase kickback: a controlled-U acting on an eigenstate of U moves the eigenvalue phase onto the control qubit.

Relevant formulae. C-$U(\alpha|0\rangle + \beta|1\rangle)|u\rangle = (\alpha|0\rangle + \beta e^{i\phi}|1\rangle)|u\rangle$ if $U|u\rangle = e^{i\phi}|u\rangle$.

Step-by-step explanation. Linearity: C-U $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle|u\rangle + |1\rangle U|u\rangle) = \frac{1}{\sqrt{2}}\left(|0\rangle|u\rangle + e^{i\phi}|1\rangle|u\rangle\right)$. Because $U|u\rangle$ is $|u\rangle$ times a number, the number can be attached to the control: $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle) \otimes |u\rangle$. This is **phase kickback**: the eigenvalue becomes a *relative* phase of the control, the target is unchanged and no entanglement is created. Phase estimation reads these kicked-back phases with the inverse QFT.

Why the other options are incorrect.

(A) "The phase stays on the target": $e^{i\phi}$ multiplies only the $|1\rangle$-control branch, so it is a *relative* phase of the control, not a global factor.

(B) $U^\dagger$ appears nowhere and, since $|u\rangle$ is an eigenstate, no entanglement is created.

(C) $(e^{i\phi}|0\rangle + |1\rangle)$ puts the phase on the wrong branch: the $|0\rangle$ branch is untouched by a controlled gate.

Key takeaway. Eigenvalue phases are read off the control register — the core of phase estimation and hence of HHL.

Q.27 SINGLE CORRECT · Inverse QFT of a phase state

The inverse QFT on two qubits is applied to $|\psi\rangle = \frac{1}{2}\sum_{y=0}^{3} e^{2\pi i\, 3y/4}|y\rangle$. The result is

(A) $\frac{1}{2}\sum_y |y\rangle$ **(B)** $|10\rangle$ **(C)** $|01\rangle$ **(D)** ✓ $|11\rangle$

Answer: (D) Detailed solution

Correct answer. (D) $|11\rangle$

Concepts required. The QFT maps $|x\rangle$ to a state whose phases advance by $2\pi x/N$ per step; the inverse QFT reads x back.

Relevant formulae. $\mathrm{QFT}|x\rangle = \frac{1}{\sqrt{N}}\sum_y e^{2\pi i xy/N}|y\rangle$; $\mathrm{QFT}^{-1}\mathrm{QFT}|x\rangle = |x\rangle$.

Step-by-step explanation. By definition $\mathrm{QFT}|x\rangle = \frac{1}{2}\sum_y e^{2\pi i\, xy/4}|y\rangle$; for $x = 3$ this is exactly $|\psi\rangle$. Hence $\mathrm{QFT}^{-1}|\psi\rangle = |3\rangle = |11\rangle$. Explicitly the amplitudes are $\frac{1}{2}(1, -i, -1, i)$: the phase advances by $3 \cdot \frac{2\pi}{4}$ per step. (The equivalent reading $e^{2\pi i\, 3y/4} = e^{-2\pi i\, y/4}$ means "-1", which is 3 modulo 4 — not 1.)

Why the other options are incorrect.

(A) The inverse QFT outputs the uniform superposition only for the input $|00\rangle$ ($\mathrm{QFT}^{-1}|00\rangle = \frac{1}{2}\sum_y |y\rangle$); a phase gradient maps to a single basis state.

(B) $|10\rangle = |2\rangle$ would need amplitudes $\frac{1}{2}(1, -1, 1, -1)$.

(C) $|01\rangle$ reads $e^{2\pi i\, 3y/4} = e^{-2\pi i y/4}$ as "$+1$"; it is $-1 \equiv 3 \pmod 4$.

Key takeaway. A linear phase gradient $e^{2\pi i xy/N}$ is $\mathrm{QFT}|x\rangle$; the inverse QFT turns it into $|x\rangle$.

Q.28 SINGLE CORRECT · QPE with an exact 3-bit phase

Phase estimation with $n_c = 3$ clock qubits is run on an eigenstate with $\varphi = 0.375$. The measured clock bit string (c_0 = most significant bit) is

(A) 100 with probability 1 **(B)** ✓ 011 with probability 1

(C) 011 with probability ≈ 0.81 **(D)** 110 with probability 1

Answer: (B) Detailed solution

Correct answer. (B) 011 with probability 1

Concepts required. QPE is exact when $\varphi\, 2^{n_c}$ is an integer: the clock ends in a single basis state.

Relevant formulae. $k = \varphi\, 2^{n_c}$; $0.375 = 0.011_2$; $\tilde{\varphi} = k/2^{n_c}$.

Step-by-step explanation. $\varphi = 0.375 = \frac{3}{8} = 0.011_2$ is an exact 3-bit binary fraction. After

the controlled powers the clock holds $\frac{1}{\sqrt{8}}\sum_k e^{2\pi i\,3k/8}\,|k\rangle = \mathrm{QFT}\,|3\rangle$, so the inverse QFT returns $|3\rangle = |011\rangle$ with certainty and $\tilde{\varphi} = 3/8$. Probabilities below 1 (like the $4/\pi^2 \approx 0.405$ bound) occur only when $\varphi\,2^{n_c}$ is *not* an integer. 110 is the bit-reversed string (a classic qubit-ordering slip); $100 = 4$ would mean $\varphi = 0.5$.

Why the other options are incorrect.

(A) $100 = 4$ corresponds to $\varphi = 0.5$.

(C) 0.81 (or $4/\pi^2$-type values) apply only to non-representable phases; here the phase is exact.

(D) 110 reverses the bit order: 0.011_2 read backwards.

Key takeaway. Exact binary phase ⇒ deterministic QPE outcome. HHL is exact when every $\lambda_j t 2^{n_c}/2\pi$ is an integer.

Q.29 SINGLE CORRECT · QPE with a non-representable phase

Phase estimation with $n_c = 3$ clock qubits is run on an eigenstate with $\varphi = \frac{1}{3}$. The most likely outcome k and its probability are

(A) ✓ $k = 3;\ \approx 0.69$ **(B)** $k = 3;\ \approx 0.41$ **(C)** $k = 3;\ 1$ **(D)** $k = 2;\ \approx 0.69$

Answer: (A) Detailed solution

Correct answer. (A) $k = 3;\ \approx 0.69$

Concepts required. For non-representable phases QPE outputs a distribution peaked at the nearest integer(s) to $\varphi\,2^{n_c}$.

Relevant formulae. $P(k) = \frac{\sin^2(\pi 2^{n_c}\delta)}{2^{2n_c}\sin^2(\pi\delta)}$, $\delta = \varphi - k/2^{n_c}$; worst case $P_{\text{best}} \geq 4/\pi^2$.

Step-by-step explanation. $\varphi\,2^{n_c} = 8/3 \approx 2.667$ is not an integer, so the outcome is random, peaked at the nearest integer $k = 3$. For outcome k, with $\delta = \varphi - k/8$:

$$P(k) = \left|\frac{1}{8}\sum_{m=0}^{7} e^{2\pi i\, m\delta}\right|^2 = \frac{\sin^2(8\pi\delta)}{64\sin^2(\pi\delta)}.$$

For $k = 3$: $\delta = \frac{1}{3} - \frac{3}{8} = -\frac{1}{24}$, $\sin^2(\pi/3) = 0.75$, $64\sin^2(\pi/24) = 64 \times 0.017037 = 1.0904$, so $P(3) \approx 0.688$. (For $k = 2$: $\delta = \frac{1}{12}$, $P(2) = \frac{\sin^2(2\pi/3)}{64\sin^2(\pi/12)} = \frac{0.75}{4.287} \approx 0.175$.) The bound $4/\pi^2 \approx 0.405$ is only a worst-case guarantee. In HHL such rounding of $\lambda t 2^{n_c}/2\pi$ is the main source of error.

Why the other options are incorrect.

(B) $0.41 \approx 4/\pi^2$ is the worst-case lower bound (reached when $\varphi 2^{n_c}$ is half-way between integers), not the value here.

(C) Probability 1 requires 8/3 to be an integer; it is not.

(D) $k = 2$ is the second-nearest value ($|8/3 - 2| = 0.67 > |8/3 - 3| = 0.33$); its probability is only 0.175.

Key takeaway. Rounding in QPE spreads probability over neighbouring clock values; in HHL this means applying slightly wrong $1/\lambda$ factors.

Q.30 Statement I/II · Unitarity vs. measurement

Statement I: A unitary gate preserves the norm of every state vector.

Statement II: A projective measurement is reversible: the pre-measurement state can always be recovered from the post-measurement state.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. **(C)** Statement I is true, but Statement II is false
Concepts required. Unitary evolution is reversible and norm-preserving; projective measurement is irreversible.
Relevant formulae. $U^\dagger U = I \Rightarrow \|U\psi\| = \|\psi\|$; measurement: $|\psi\rangle \to \frac{P_k|\psi\rangle}{\|P_k|\psi\rangle\|}$.
Step-by-step explanation. **I is true:** $\|U\psi\|^2 = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle$. **II is false:** measurement maps $\sum_k a_k |k\rangle$ to a single $|k\rangle$; all the other amplitudes are lost and different states (e.g. $|+\rangle$ and $|-\rangle$) give the same post-measurement states. This irreversibility is why HHL's ancilla measurement must be the *last* step on the system, and why a failed run (ancilla = 0) cannot be repaired — it must be repeated. Answer: (C).
Why the other options are incorrect.

(A) accepts II, but different states ($|+\rangle$, $|-\rangle$) give the same post-measurement states, so measurement cannot be undone.

(B) rejects I, but unitarity is exactly norm preservation.

(D) rejects I and accepts II — both reversed.

Key takeaway. Measure last. In HHL the ancilla measurement ends the computation; a failure means starting again.

Q.31 Assertion–Reason · Global vs. relative phase

Assertion (A): The states $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ give identical statistics for every measurement.

Reason (R): The relative phase between the components of a superposition affects interference and hence measurement results.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) ✓ Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (B) Detailed solution

Correct answer. **(B)** Both A and R are true, but R is *not* the correct explanation of A
Concepts required. Global phases are unobservable; relative phases change interference.
Relevant formulae. $|e^{i\alpha}\langle m|\psi\rangle|^2 = |\langle m|\psi\rangle|^2$; $H|\pm\rangle = |0\rangle, |1\rangle$.
Step-by-step explanation. **A is true:** every probability has the form $|\langle m|\psi\rangle|^2$ and $|e^{i\alpha}\langle m|\psi\rangle|^2 = |\langle m|\psi\rangle|^2$ — a *global* phase is unobservable. **R is true:** e.g. $|+\rangle$ and $|-\rangle$ differ only by a relative phase, yet H maps them to $|0\rangle$ and $|1\rangle$. R does not explain A: A is about a phase common to all components, R about phases *between* components. Answer: (B). (In HHL, the sign of C/λ_j is a relative phase between eigencomponents and therefore matters.)
Why the other options are incorrect.

(A) claims R explains A, but R concerns relative phases and A global ones; A follows from $|e^{i\alpha}| = 1$.

(C) rejects R, but $|+\rangle$ and $|-\rangle$ differ only by a relative phase and are perfectly distinguishable.

(D) rejects A, but every probability is unchanged by a common phase.

Key takeaway. Global phase: irrelevant. Relative phase (e.g. the sign of $1/\lambda_j$ in HHL): essential.

Q.32 SINGLE CORRECT · Controlled-R_Y and ancilla probability

A controlled-$R_Y(\theta)$ with $\theta = 2\arcsin\frac{1}{2}$ acts on control $|1\rangle$ and target (ancilla) $|0\rangle$. The probability of then measuring the ancilla as 1 is

(A) ✓ $\frac{1}{4}$ **(B)** $\frac{3}{4}$ **(C)** $\frac{1}{16}$ **(D)** $\frac{1}{2}$

Answer: (A) Detailed solution

Correct answer. (A) $\frac{1}{4}$

Concepts required. A controlled rotation acts only when the control is $|1\rangle$; then the ancilla amplitude on $|1\rangle$ is $\sin\frac{\theta}{2}$.

Relevant formulae. $\theta = 2\arcsin c \Rightarrow R_Y(\theta)|0\rangle = \sqrt{1-c^2}\,|0\rangle + c\,|1\rangle$, $P(1) = c^2$.

Step-by-step explanation. $\theta = 2\arcsin\frac{1}{2} = 2\cdot\frac{\pi}{6} = \frac{\pi}{3}$. The control is $|1\rangle$, so the rotation acts: $R_Y(\pi/3)|0\rangle = \cos\frac{\pi}{6}|0\rangle + \sin\frac{\pi}{6}|1\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$. Hence $P(1) = (\frac{1}{2})^2 = \frac{1}{4}$. In HHL this is the eigenvalue inversion for $C/\lambda = \frac{1}{2}$ (e.g. $C = 1$, $\lambda = 2$): the amplitude is C/λ and the success probability of that component is $(C/\lambda)^2$.

Why the other options are incorrect.

(B) $\frac{3}{4}$ is the probability of the ancilla staying in $|0\rangle$.

(C) $\frac{1}{16}$ squares twice (c^4).

(D) $\frac{1}{2}$ is the amplitude c, not the probability c^2.

Key takeaway. $\theta = 2\arcsin c$ loads amplitude c and probability c^2 onto $|1\rangle$ — HHL's eigenvalue inversion.

Q.33 SINGLE CORRECT · Selecting a clock value with X gates

A 3-qubit clock register $c_0c_1c_2$ (c_0 = most significant bit) controls a multi-controlled R_Y that fires only when all controls are 1. To make it act only on the clock value $k = 2$, the code places X gates before and after the rotation. Which X gates are needed?

(A) X on c_1 and c_2, before and after (4 gates)

(B) X on c_1 only, before and after (2 gates)

(C) ✓ X on c_0 and c_2, before and after (4 gates)

(D) none: $k = 2$ is selected by the rotation angle

Answer: (C) Detailed solution

Correct answer. (C) X on c_0 and c_2, before and after (4 gates)

Concepts required. Multi-controlled gates fire on all-ones; to condition on another pattern, flip the controls that should be 0 (and flip them back).

Relevant formulae. $k = \sum_i c_i 2^{n_c-1-i}$ (c_0 = MSB); flip every $c_i = 0$; gates = 2×(number of zero bits).

Step-by-step explanation. $k = 2 = 010_2$: $c_0 = 0$, $c_1 = 1$, $c_2 = 0$. A multi-controlled gate fires on $|111\rangle$, so every control that must be 0 is flipped by X first: X on c_0 and c_2 maps $|010\rangle \to |111\rangle$. After the rotation the same X gates restore the clock. Total $2 \times 2 = 4$ X gates. This is exactly the `zero_bits` loop in `HHL.ipynb`: `kernel.x(clock[i])` for every bit i of k that is 0.

Why the other options are incorrect.

(A) X on c_1 and c_2 selects $100_2 = 4$; c_0 is the most significant bit.

(B) X on c_1 flips the bit that is already 1 and selects $k = 101_2 = 5$ instead of 2.

(D) The rotation angle does not select anything: without X gates the gate fires only for $k = 7$.

Key takeaway. X–controlled gate–X "polarity" trick: one multi-controlled rotation per clock value in `HHL.ipynb`.

Q.34 SINGLE CORRECT · Post-selection on an ancilla

An ancilla a and a system register are in $\frac{1}{2}|0\rangle_a|\psi_0\rangle + \frac{\sqrt{3}}{2}|1\rangle_a|\psi_1\rangle$ ($|\psi_0\rangle, |\psi_1\rangle$ normalised). The ancilla is measured and the result 1 is kept (post-selection). The probability of this and the resulting system state are

(A) ✓ $\frac{3}{4}$; $|\psi_1\rangle$

(B) $\frac{\sqrt{3}}{2}$; $|\psi_1\rangle$

(C) $\frac{1}{4}$; $|\psi_0\rangle$

(D) $\frac{3}{4}$; $\frac{1}{\sqrt{2}}(|\psi_0\rangle + |\psi_1\rangle)$

Answer: (A) Detailed solution

Correct answer. **(A)** $\frac{3}{4}$; $|\psi_1\rangle$

Concepts required. Post-selection keeps only the runs with the desired measurement outcome; the kept state is the renormalised branch.

Relevant formulae. $P(1) = \|\Pi_1|\Psi\rangle\|^2$; $|\Psi'\rangle = \Pi_1|\Psi\rangle / \sqrt{P(1)}$.

Step-by-step explanation. $P(a=1) = \left\|\frac{\sqrt{3}}{2}|\psi_1\rangle\right\|^2 = \frac{3}{4}$. After the measurement only the $a=1$ branch survives and is renormalised: $\frac{\frac{\sqrt{3}}{2}|\psi_1\rangle}{\sqrt{3/4}} = |\psi_1\rangle$. HHL works in exactly this way: the ancilla-1 branch holds $\propto A^{-1}|b\rangle$, the ancilla-0 branch is discarded, and the probability of keeping it is the success probability.

Why the other options are incorrect.

(B) $\frac{\sqrt{3}}{2}$ is the amplitude of the branch, not its probability.

(C) $\frac{1}{4}$ and $|\psi_0\rangle$ describe the other outcome ($a=0$), which HHL discards.

(D) The $|\psi_0\rangle$ branch is removed by the measurement; no superposition of both branches survives.

Key takeaway. Post-selection probability = squared norm of the kept branch; in HHL it is $P_{\text{success}} = C^2\|A^{-1}|b\rangle\|^2$.

Q.35 SINGLE CORRECT · Inverse of a circuit

A circuit applies U_1, then U_2, then U_3 (all unitary). The circuit that undoes it applies

(A) $U_1^\dagger$, then $U_2^\dagger$, then $U_3^\dagger$

(B) U_3, then U_2, then U_1

(C) $U_1^\dagger U_2^\dagger U_3^\dagger$ applied as a single gate in the same order as the original

(D) ✓ $U_3^\dagger$, then $U_2^\dagger$, then $U_1^\dagger$

Answer: (D) Detailed solution

Correct answer. **(D)** $U_3^\dagger$, then $U_2^\dagger$, then $U_1^\dagger$

Concepts required. The inverse of a product reverses the order: $(U_3U_2U_1)^{-1} = U_1^{-1}U_2^{-1}U_3^{-1}$.

Relevant formulae. $(AB)^\dagger = B^\dagger A^\dagger$; a circuit "$U_1$ then U_2" is the matrix U_2U_1.

Step-by-step explanation. The circuit implements the product $W = U_3U_2U_1$ (the first gate acts first, so it stands on the right). $W^{-1} = W^\dagger = U_1^\dagger U_2^\dagger U_3^\dagger$; read right to left this means: apply $U_3^\dagger$ first, then $U_2^\dagger$, then $U_1^\dagger$. Reverse the order *and* invert every gate. HHL's uncomputation does exactly this with phase estimation: the QFT (inverse of the inverse QFT) first, then the controlled U^{-2^j} in reverse order, then the Hadamards.

Why the other options are incorrect.

(A) Inverting the gates without reversing the order gives $U_3^\dagger U_2^\dagger U_1^\dagger$, which is not W^{-1} when the gates do not commute.

(B) Reversing the order without inverting the gates gives $U_1U_2U_3$, not the inverse.

(C) Combining the daggers into one gate in the original order is the same mistake as the first option.

Key takeaway. Undo = reverse order and dagger every gate. HHL's uncomputation is QPE run backwards.

Q.36 Matching lists · Gate counts of a 4-qubit QFT

The textbook QFT on $n = 4$ qubits (as used, inverted, on the clock register of the default HHL circuit). Match each quantity (List-I) with its value (List-II).

List-I		List-II	
(P)	number of Hadamard gates	**(1)**	2
(Q)	number of controlled-phase gates	**(2)**	$\pi/8$
(R)	number of SWAP gates	**(3)**	6
(S)	magnitude of the smallest controlled-phase angle	**(4)**	4

(A) P→3, Q→4, R→1, S→2 **(B)** P→4, Q→1, R→3, S→2

(C) P→4, Q→3, R→2, S→1 **(D)** ✓ P→4, Q→3, R→1, S→2

Answer: (D) Detailed solution

Correct answer. (D) P→4, Q→3, R→1, S→2

Concepts required. The n-qubit QFT circuit: n Hadamards, $n(n-1)/2$ controlled phases R_k, $\lfloor n/2 \rfloor$ SWAPs.

Relevant formulae. $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, $k = 2, \ldots, n$; smallest angle $2\pi/2^n$.

Step-by-step explanation.

- P: one Hadamard per qubit: 4 (4).
- Q: $n(n-1)/2 = 4 \cdot 3/2 = 6$ controlled phases (3).
- R: $\lfloor n/2 \rfloor = 2$ SWAPs to reverse the qubit order (1).
- S: angles $2\pi/2^k$ for $k = 2, \ldots, n$; the smallest is $2\pi/2^4 = \pi/8$ (2).

Consistency with `HHL.ipynb`: the recorded statistics show 12 `cr1` and 4 `swap` gates (inverse QFT + QFT for uncomputation: 2×6 and 2×2) and 16 `h` gates (4 initial + 4 + 4 in the two QFTs + 4 final).

Why the other options are incorrect.

(A) P→3, Q→4: there is one Hadamard per qubit (4), and $6 = 4 \cdot 3/2$ controlled phases.

(B) Q→1, R→3 exchanges the counts: 6 controlled phases and 2 SWAPs, not 2 and 6.

(C) R→2, S→1 gives $\pi/8$ SWAPs and an angle of 2 — units mixed up.

Key takeaway. QFT cost is $O(n^2)$ gates; in HHL it is negligible compared with the controlled e^{iAt2^j}.

Q.37 SINGLE CORRECT · Uniform superposition

A Hadamard is applied to each of four qubits in $|0000\rangle$. The amplitude of $|1011\rangle$ is

(A) $\frac{1}{16}$ **(B)** ✓ $\frac{1}{4}$ **(C)** $-\frac{1}{4}$ **(D)** $\frac{1}{\sqrt{8}}$

Answer: (B) Detailed solution

Correct answer. (B) $\frac{1}{4}$

Concepts required. $H^{\otimes n}|0\ldots0\rangle$ is the uniform superposition with all amplitudes $2^{-n/2}$.

Relevant formulae. $H^{\otimes n}|x\rangle = 2^{-n/2}\sum_y (-1)^{x\cdot y}|y\rangle$; for $x=0$ all signs are $+$.

Step-by-step explanation. $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$ for each qubit, so $H^{\otimes 4}|0000\rangle = \frac{1}{(\sqrt{2})^4}\sum_{y=0}^{15}|y\rangle = \frac{1}{4}\sum_y|y\rangle$. Every one of the 16 basis states has amplitude $\frac{1}{4}$ (probability $\frac{1}{16}$), with a $+$ sign because the input is $|0\ldots0\rangle$ (signs $(-1)^{x\cdot y}$ appear only for inputs $x\neq 0$). This is the first step of phase estimation on the HHL clock register.

Why the other options are incorrect.

(A) $\frac{1}{16}$ is the probability, not the amplitude.

(C) $-\frac{1}{4}$: signs $(-1)^{x\cdot y}$ appear only for inputs $x\neq 0$.

(D) $\frac{1}{\sqrt{8}}$ corresponds to 3 qubits.

Key takeaway. The clock register starts in the uniform superposition; QPE then writes phases into it.

Q.38 STATEMENT I/II · Product states, reduced states

Statement I: Every two-qubit pure state can be written as $|a\rangle\otimes|b\rangle$ for suitable single-qubit states $|a\rangle$, $|b\rangle$.

Statement II: For the Bell state $\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle)$ the reduced state of each qubit is the maximally mixed state $\frac{1}{2}I$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

Correct answer. (D) Statement I is false, but Statement II is true

Concepts required. Entangled states cannot be factorised; their subsystems are in mixed states.

Relevant formulae. $\rho_A = \mathrm{Tr}_B|\psi\rangle\langle\psi|$; Bell state: $\rho_A = \frac{1}{2}I$; product $\iff$ rank-1 coefficient matrix.

Step-by-step explanation. **I is false:** product states form a small subset; the Bell state is a counter-example (its coefficient matrix $\frac{1}{\sqrt{2}}I$ has rank 2). **II is true:** tracing out the second qubit gives $\rho_A = \frac{1}{2}(|0\rangle\langle 0|+|1\rangle\langle 1|) = \frac{1}{2}I$; a measurement of one qubit alone gives 0 or 1 with probability $\frac{1}{2}$ each. In HHL, if the clock were *not* uncomputed, the reduced state of the system register would be mixed like this instead of the pure $|x\rangle$. Answer: (D).

Why the other options are incorrect.

(A) accepts I, but entangled states (e.g. Bell states) are not products.

(B) rejects II, but the reduced state of a Bell state is $\frac{1}{2}I$.

(C) accepts I and rejects II — both reversed.

Key takeaway. Leaving the clock entangled with the system would turn HHL's pure output $|x\rangle$ into a mixture.

Paragraph for Questions 39 and 40

A unitary U has the eigenstate $|u\rangle$ with $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $\varphi = \frac{1}{4}$. Phase estimation uses $n_c = 2$ clock qubits c_0c_1 (c_0 = most significant bit). Both start in $|0\rangle$ and receive a Hadamard; c_1 (weight 2^0) controls U and c_0 (weight 2^1) controls U^2, so that the state becomes $\frac{1}{2}\sum_{k=0}^{3} e^{2\pi i\varphi k}|k\rangle|u\rangle$. Finally the inverse QFT is applied to the clock and the clock is measured.

Q.39 PARAGRAPH BASED · QPE trace: phase state

Before the inverse QFT, the amplitude of the clock state $|11\rangle$ (with $|u\rangle$ factored out) is

(A) $\frac{1}{2}$ **(B)** ✓ $-\frac{i}{2}$ **(C)** $-\frac{1}{2}$ **(D)** $\frac{i}{2}$

Answer: (B) Detailed solution

Correct answer. **(B)** $-\frac{i}{2}$

Concepts required. After the controlled powers, clock value k carries the phase $e^{2\pi i\varphi k}$.

Relevant formulae. $\frac{1}{\sqrt{2^{n_c}}}\sum_k e^{2\pi i\varphi k}|k\rangle$; $e^{2\pi i\cdot 3/4} = e^{3\pi i/2} = -i$.

Step-by-step explanation. The clock amplitudes are $\frac{1}{2}e^{2\pi i\varphi k}$ with $\varphi = \frac{1}{4}$, i.e. $\frac{1}{2}i^k$. For $|11\rangle$, $k = 3$: $\frac{1}{2}i^3 = \frac{1}{2}(-i) = -\frac{i}{2}$. The four amplitudes are $\frac{1}{2}(1, i, -1, -i)$: each controlled power adds the phase $2\pi\varphi\cdot 2^j$ to the clock qubit of weight 2^j (phase kickback, Q.26).

Why the other options are incorrect.

(A) $\frac{1}{2}$ is the amplitude of $|00\rangle$; it ignores the phase.

(C) $-\frac{1}{2}$ is the amplitude of $|10\rangle$ ($k = 2$).

(D) $\frac{i}{2}$ is the amplitude of $|01\rangle$ ($k = 1$), not of $|11\rangle$.

Key takeaway. Clock value k accumulates k times the eigenphase — a phase gradient, i.e. a Fourier state.

Q.40 PARAGRAPH BASED · QPE trace: outcome

After the inverse QFT the clock measurement gives

(A) ✓ 01 with probability 1, so $\tilde{\varphi} = \frac{1}{4}$ **(B)** 01 or 11 with probability $\frac{1}{2}$ each

(C) 11 with probability 1, so $\tilde{\varphi} = \frac{3}{4}$ **(D)** 10 with probability 1, so $\tilde{\varphi} = \frac{1}{2}$

Answer: (A) Detailed solution

Correct answer. **(A)** 01 with probability 1, so $\tilde{\varphi} = \frac{1}{4}$

Concepts required. If $\varphi 2^{n_c}$ is an integer k, the clock is exactly $\mathrm{QFT}|k\rangle$ and the inverse QFT returns $|k\rangle$.

Relevant formulae. $\frac{1}{2}\sum_y e^{2\pi i\,ky/4}|y\rangle = \mathrm{QFT}|k\rangle$; $\tilde{\varphi} = k/2^{n_c}$.

Step-by-step explanation. The clock state $\frac{1}{2}\sum_k e^{2\pi i\,k/4}|k\rangle$ is $\mathrm{QFT}|1\rangle$ (definition with $N = 4$, $x = 1$). Hence the inverse QFT gives $|1\rangle = |01\rangle$ with certainty, and $\tilde{\varphi} = 1/2^{n_c} = 1/4$ — exact, because $\varphi\cdot 2^{n_c} = 1$ is an integer. In HHL the same mechanism writes $k = \lambda t 2^{n_c}/2\pi$ into the clock, one value per eigencomponent of $|b\rangle$.

Why the other options are incorrect.

(B) An exactly representable phase gives a deterministic outcome, not a 50/50 split.

(C) 11 ($\tilde{\varphi} = \frac{3}{4}$) corresponds to the conjugate phase $e^{-2\pi i/4}$.

(D) 10 ($\tilde{\varphi} = \frac{1}{2}$) would need the amplitudes $\frac{1}{2}(1, -1, 1, -1)$.

Key takeaway. Exact phase, exact readout: this is why the default HHL example (integer clock values) has fidelity 1.

SECTION C · HHL Algorithm Fundamentals Q.41–60

Q.41 SINGLE CORRECT · What HHL outputs

For a Hermitian A and a prepared $|b\rangle$, a successful run of the HHL algorithm produces

(A) the eigenvalues of A as a classical list

(B) the classical list of all d numbers $x_1, \ldots, x_d$

(C) ✓ the quantum state $|x\rangle \propto A^{-1}|b\rangle$ in the system register

(D) the matrix A^{-1} stored in the clock register

Answer: (C) Detailed solution

Correct answer. **(C)** the quantum state $|x\rangle \propto A^{-1}|b\rangle$ in the system register

Concepts required. HHL is a state-preparation algorithm: its output is a quantum state whose amplitudes are proportional to $\vec{x}$.

Relevant formulae. $|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|} = \frac{\sum_j (\beta_j/\lambda_j)|u_j\rangle}{\sqrt{\sum_j |\beta_j|^2/\lambda_j^2}}$.

Step-by-step explanation. HHL transforms $|b\rangle = \sum_j \beta_j |u_j\rangle$ into $|x\rangle = \frac{\sum_j (\beta_j/\lambda_j)|u_j\rangle}{\|\cdot\|}$, a normalised quantum state whose amplitudes are proportional to the solution vector. The d numbers are *encoded* in $\log_2 d$ qubits; reading them all out needs $\Omega(d)$ repetitions (Section D), so HHL is useful when only a property of $\vec{x}$ (an expectation value, an overlap) is needed. The eigenvalues are used only temporarily inside the circuit and are uncomputed; A^{-1} is never stored.

Why the other options are incorrect.

(A) The eigenvalues are written into the clock only coherently and are erased again; they are not measured or output.

(B) A classical list of d numbers would need $\Omega(d)$ measurements; one run yields one sample.

(D) A^{-1} (d^2 numbers) is never formed; the clock holds eigenvalue estimates temporarily and is uncomputed.

Key takeaway. HHL outputs $|x\rangle$, not $\vec{x}$. Everything else about its usefulness follows from this.

Q.42 SINGLE CORRECT · State after phase estimation

With $|b\rangle = \sum_j \beta_j |u_j\rangle$ and exactly representable clock values k_j, the state after phase estimation (clock $\otimes$ system $\otimes$ ancilla) is

(A) $|0\ldots0\rangle\,|x\rangle\,|1\rangle$

(B) $\left(\frac{1}{\sqrt{d}}\sum_j |k_j\rangle\right) \otimes |b\rangle \otimes |0\rangle$

(C) $\sum_j \frac{\beta_j}{\lambda_j} |k_j\rangle\,|u_j\rangle\,|0\rangle$

(D) ✓ $\sum_j \beta_j |k_j\rangle\,|u_j\rangle\,|0\rangle$

Answer: (D) Detailed solution

Correct answer. **(D)** $\sum_j \beta_j |k_j\rangle\,|u_j\rangle\,|0\rangle$

Concepts required. Phase estimation acts eigencomponent by eigencomponent (linearity) and writes each eigenvalue into the clock.

Relevant formulae. $|0\rangle\,|u_j\rangle \xrightarrow{\text{QPE}} |k_j\rangle\,|u_j\rangle$, $k_j = \lambda_j t 2^{n_c}/2\pi$; $|b\rangle = \sum_j \beta_j |u_j\rangle$.

Step-by-step explanation. Phase estimation is linear. On each eigencomponent $|u_j\rangle$, $U = e^{iAt}$ acts as the phase $e^{i\lambda_j t} = e^{2\pi i \varphi_j}$ with $\varphi_j = \lambda_j t/2\pi$, and QPE writes $k_j = \varphi_j 2^{n_c}$ into the clock: $|0\rangle\,|u_j\rangle \mapsto |k_j\rangle\,|u_j\rangle$. Summing with weights β_j:

$$|0\rangle\,|b\rangle\,|0\rangle = \sum_j \beta_j |0\rangle\,|u_j\rangle\,|0\rangle \;\longmapsto\; \sum_j \beta_j |k_j\rangle\,|u_j\rangle\,|0\rangle\,.$$

The clock is now *entangled* with the system; no factor $1/\lambda_j$ has appeared yet — that is the job of

the controlled rotation.

Why the other options are incorrect.

(A) $|0\dots0\rangle\,|x\rangle\,|1\rangle$ is the (ideal) state at the very end, after rotation, uncomputation and post-selection.

(B) A product state ignores that each eigenvalue is correlated with its own eigenvector: QPE entangles clock and system.

(C) The factors $1/\lambda_j$ appear only after the controlled rotation and post-selection, not after QPE.

Key takeaway. After QPE: $\sum_j \beta_j\,|k_j\rangle\,|u_j\rangle$ — eigenvalues sit next to their eigenvectors, ready to be inverted.

Q.43 SINGLE CORRECT · Eigenvalue → clock bit string

In the default notebook run ($n_c = 4$, $t = \pi/8$), the eigenvalue $\lambda = 3$ is written into the clock register as the bit string

(A) 0010 **(B)** 0110 **(C)** ✓ 0011 **(D)** 1100

Answer: (C) Detailed solution

Correct answer. (C) 0011

Concepts required. The clock value is the eigenphase in units of $2\pi/2^{n_c}$, written in binary with the most significant bit first.

Relevant formulae. $k = \dfrac{\lambda t 2^{n_c}}{2\pi}$; default $t = \pi/8$, $n_c = 4 \Rightarrow k = \lambda$.

Step-by-step explanation. $k = \dfrac{\lambda t\,2^{n_c}}{2\pi} = \dfrac{3 \cdot \frac{\pi}{8} \cdot 16}{2\pi} = \dfrac{6\pi}{2\pi} = 3$. With $n_c = 4$ bits and the most significant bit first, $3 = 0011_2$. In fact $t = \pi/8$ was chosen ("auto") so that $k = \lambda$ for every eigenvalue: $1, 2, 3, 4 \to 0001, 0010, 0011, 0100$, all exact, which is why the recorded fidelity is 1. 1100 is the bit-reversed string; $0110 = 6$ comes from dividing by π instead of 2π.

Why the other options are incorrect.

(A) $0010 = 2$ is the clock value of $\lambda = 2$.

(B) $0110 = 6$ divides by π instead of 2π.

(D) 1100 is 0011 written in reverse bit order (least significant bit first).

Key takeaway. Choose t so that every $\lambda t 2^{n_c}/2\pi$ is an integer and QPE becomes exact.

Q.44 SINGLE CORRECT · Constant C and rotation angle

For the default notebook run ($n_c = 4$, $t = \pi/8$) the constant C and the rotation angle θ applied for the clock value of $\lambda = 2$ are

(A) $C = \frac{1}{16}$; $\theta = 2\arcsin\frac{1}{32}$ **(B)** ✓ $C = 1$; $\theta = \pi/3$

(C) $C = 1$; $\theta = \pi/6$ **(D)** $C = \pi/8$; $\theta = 2\arcsin(\pi/16)$

Answer: (B) Detailed solution

Correct answer. (B) $C = 1$; $\theta = \pi/3$

Concepts required. C is the eigenvalue represented by clock value 1; the rotation angle loads the amplitude $C/\tilde{\lambda}$.

Relevant formulae. $C = \dfrac{2\pi}{2^{n_c}t}$, $\tilde{\lambda}_k = kC$, $\theta_k = 2\arcsin\dfrac{C}{\tilde{\lambda}_k}$.

Step-by-step explanation. $C = \dfrac{2\pi}{2^{n_c}t} = \dfrac{2\pi}{16 \cdot \pi/8} = \dfrac{2\pi}{2\pi} = 1$ — the eigenvalue that maps to clock value 1. For $\lambda = 2$: $\theta = 2\arcsin\frac{C}{\lambda} = 2\arcsin\frac{1}{2} = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$, so the ancilla becomes $\cos\frac{\pi}{6}\,|0\rangle + \sin\frac{\pi}{6}\,|1\rangle = \frac{\sqrt{3}}{2}\,|0\rangle + \frac{1}{2}\,|1\rangle$: amplitude $C/\lambda = \frac{1}{2}$ on $|1\rangle$. (Note $R_Y(\theta)$ rotates by $\theta/2$ in

amplitude, hence the factor 2.)

Why the other options are incorrect.

(A) $C = \frac{1}{16}$ uses $1/2^{n_c}$ and forgets the factor $2\pi/t$.

(C) $\theta = \pi/6$ forgets the factor 2: $\arcsin\frac{1}{2} = \frac{\pi}{6}$ is the half-angle.

(D) $C = \pi/8$ confuses C with the evolution time t.

Key takeaway. Default run: $C = 1$, so the ancilla amplitude for eigenvalue λ is simply $1/\lambda$.

Q.45 SINGLE CORRECT · Ancilla amplitude C/λ

With $C = 1$, the eigencomponent with $\lambda = 4$ leaves the ancilla in $\sqrt{1 - C^2/\lambda^2}\,|0\rangle + \frac{C}{\lambda}|1\rangle$. For this component the amplitude on $|1\rangle$ and the probability of measuring the ancilla as 1 are

(A) 4 and 1 **(B)** $\frac{1}{4}$ and $\frac{1}{4}$ **(C)** $\frac{1}{2}$ and $\frac{1}{4}$ **(D)** ✓ $\frac{1}{4}$ and $\frac{1}{16}$

Answer: (D) Detailed solution

Correct answer. (D) $\frac{1}{4}$ and $\frac{1}{16}$

Concepts required. Eigenvalue inversion: the ancilla amplitude on $|1\rangle$ is C/λ; its square is the success probability of that eigencomponent.

Relevant formulae. $|0\rangle_a \to \sqrt{1 - \frac{C^2}{\lambda^2}}\,|0\rangle + \frac{C}{\lambda}|1\rangle$; $P_j = C^2/\lambda_j^2$.

Step-by-step explanation. Amplitude $C/\lambda = 1/4$; probability $|1/4|^2 = 1/16 = 0.0625$. The $|0\rangle$ amplitude is $\sqrt{15/16} = 0.968$ so the ancilla is normalised. Large eigenvalues are therefore strongly suppressed, exactly as $1/\lambda$ requires; the price is that components with large λ rarely pass post-selection. ($\frac{1}{2}$ would correspond to $1/\sqrt{\lambda}$, and an amplitude 4 is impossible — that is why $C \le |\lambda|_{\min}$ is needed.)

Why the other options are incorrect.

(A) An amplitude of 4 is impossible ($|\cdot| \le 1$); it inverts C/λ.

(B) $\frac{1}{4}$ for the probability forgets to square the amplitude.

(C) $\frac{1}{2}$ is $1/\sqrt{\lambda}$: the rotation implements $1/\lambda$, not $1/\sqrt{\lambda}$.

Key takeaway. Amplitude C/λ, probability C^2/λ^2: large eigenvalues are suppressed, small ones dominate $|x\rangle$.

Q.46 MATCHING LISTS · Roles of the registers

Match each part of the HHL circuit (List-I) with its role (List-II).

List-I		List-II	
(P)	clock register (n_c qubits)	**(1)**	is measured; outcome 1 flags a successful inversion
(Q)	system register (n_b qubits)	**(2)**	holds $\|b\rangle$ at the start and $\|x\rangle$ after success
(R)	ancilla qubit	**(3)**	holds the binary eigenvalue estimates k_j
(S)	inverse phase estimation after the rotation	**(4)**	returns the clock to $\|0\ldots0\rangle$ and removes its entanglement with the system

(A) ✓ P→3, Q→2, R→1, S→4 **(B)** P→2, Q→3, R→1, S→4

(C) P→3, Q→2, R→4, S→1 **(D)** P→3, Q→1, R→2, S→4

Answer: (A) Detailed solution

Correct answer. **(A)** P→3, Q→2, R→1, S→4
Concepts required. HHL uses three registers: clock (eigenvalues), system ($|b\rangle \to |x\rangle$) and one ancilla (success flag), plus uncomputation.
Relevant formulae. qubits $= n_c + n_b + 1$; $|0\rangle^{\otimes n_c}|b\rangle|0\rangle \to |0\rangle^{\otimes n_c}\left(\sqrt{\cdot}\,|\text{junk}\rangle|0\rangle + CA^{-1}|b\rangle|1\rangle\right)$.
Step-by-step explanation.

- P: QPE writes $k_j = \lambda_j t 2^{n_c}/2\pi$ into the clock (3).
- Q: the system starts in $|b\rangle$ (amplitude encoding) and ends in $|x\rangle$ when the run succeeds (2).
- R: the ancilla carries the amplitude $C/\tilde{\lambda}$ on $|1\rangle$; measuring 1 post-selects the A^{-1} branch (1).
- S: uncomputation maps $|k_j\rangle|u_j\rangle \to |0\rangle|u_j\rangle$ so that the eigencomponents can interfere into $|x\rangle$ (4).

Total qubits: $n_c + n_b + 1$ ($4 + 2 + 1 = 7$ in the notebook).
Why the other options are incorrect.
(B) P→2, Q→3 swaps clock and system: the eigenvalues go into the clock, $|b\rangle$ into the system.
(C) R→4, S→1 swaps the ancilla's role with uncomputation; the ancilla is the qubit that is measured.
(D) Q→1, R→2 makes the system the success flag and the ancilla the solution register.
Key takeaway. Clock = eigenvalues, system = data, ancilla = flag; uncomputation cleans up the clock.

Q.47 SINGLE CORRECT · Success probability

HHL is run with $A = \text{diag}(1, 2)$, $|b\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $C = 1$ and exactly representable eigenvalues. The success probability is

(A) 0.75 **(B)** 1 **(C)** 0.5 **(D)** ✓ 0.625

Answer: (D) Detailed solution

Correct answer. **(D)** 0.625
Concepts required. The success probability is the squared norm of the post-selected branch $C\,A^{-1}|b\rangle$.
Relevant formulae. $P_{\text{success}} = \sum_j |\beta_j|^2 \frac{C^2}{\lambda_j^2} = C^2 \| A^{-1}|b\rangle \|^2$.
Step-by-step explanation. The eigenvectors are $|0\rangle, |1\rangle$ with $\lambda = 1, 2$ and $\beta_1 = \beta_2 = \frac{1}{\sqrt{2}}$.

$$P_{\text{success}} = \sum_j |\beta_j|^2 \frac{C^2}{\lambda_j^2} = \frac{1}{2}\cdot\frac{1}{1} + \frac{1}{2}\cdot\frac{1}{4} = \frac{5}{8} = 0.625.$$

Equivalently $P = C^2 \| A^{-1}|b\rangle \|^2$ with $A^{-1}|b\rangle = \frac{1}{\sqrt{2}}(1, \frac{1}{2})$. (0.75 forgets to square C/λ.) The post-selected state is $\propto |0\rangle + \frac{1}{2}|1\rangle$, i.e. $|x\rangle = \frac{1}{\sqrt{5}}(2|0\rangle + |1\rangle)$.
Why the other options are incorrect.
(A) $0.75 = \frac{1}{2}(1 + \frac{1}{2})$ uses C/λ instead of $(C/\lambda)^2$.
(B) 1 would need $C/\lambda_j = 1$ for every component, i.e. all eigenvalues equal to C.
(C) 0.5 keeps only the $\lambda = 1$ component.
Key takeaway. $P_{\text{success}} \propto C^2/\lambda^2$: it decreases with the eigenvalues relative to C, typically like $1/\kappa^2$.

Q.48 SINGLE CORRECT · Qubit count $n_b + n_c + 1$

An 8×8 Hermitian system is solved with the notebook's circuit and $n_c = 5$ clock qubits. The total number of qubits is

(A) 6 **(B)** ✓ 9 **(C)** 8 **(D)** 14

Answer: (B) Detailed solution

Correct answer. **(B)** 9
Concepts required. Qubit counts are logarithmic in the matrix dimension.
Relevant formulae. $n_b = \log_2 d$; total $= n_c + n_b + 1$.
Step-by-step explanation. $n_b = \log_2 8 = 3$ system qubits, $n_c = 5$ clock qubits and 1 ancilla: $3 + 5 + 1 = 9$. The matrix dimension enters only logarithmically — the source of HHL's potential advantage. ($14 = 8 + 5 + 1$ confuses the dimension d with the number of qubits $\log_2 d$.)
Why the other options are incorrect.
(A) $6 = 3 + 2 + 1$ would correspond to only $n_c = 2$ clock qubits; here $n_c = 5$.
(C) 8 forgets the ancilla ($3 + 5 = 8$).
(D) 14 uses the dimension $d = 8$ instead of $\log_2 d = 3$.
Key takeaway. Qubits $= n_c + \log_2 d + 1$; the clock, not the data, usually dominates small examples.

Q.49 STATEMENT I/II · Why uncompute?

Statement I: If the clock register is not uncomputed, the post-selected system register is in general not in the pure state $|x\rangle$, because it remains entangled with the clock.

Statement II: Uncomputation can be replaced by simply measuring the clock register, with the same output $|x\rangle$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) ✓ Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. **(C)** Statement I is true, but Statement II is false
Concepts required. Uncomputation coherently erases the clock; measurement would collapse the eigencomponent superposition.
Relevant formulae. $\sum_j \frac{\beta_j}{\lambda_j} |k_j\rangle |u_j\rangle \xrightarrow{\text{QPE}^{-1}} |0\rangle \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$.
Step-by-step explanation. After the rotation and post-selection the state is $\propto \sum_j \frac{\beta_j}{\lambda_j} |k_j\rangle |u_j\rangle$. **I is true:** with distinct $|k_j\rangle$ the clock labels the eigencomponents; tracing it out leaves the mixture $\sum_j p_j |u_j\rangle\langle u_j|$, not the superposition $|x\rangle$. **II is false:** measuring the clock gives one value k_j and collapses the system to the single eigenvector $|u_j\rangle$ — the interference between eigencomponents that forms $|x\rangle$ is destroyed. Uncomputation *coherently* resets $|k_j\rangle \to |0\rangle$. Answer: (C).
Why the other options are incorrect.
(A) accepts II, but measuring the clock leaves a single eigenvector $|u_j\rangle$, not $|x\rangle$.
(B) rejects I, but an un-erased clock labels the components and the system's reduced state is mixed.
(D) rejects I and accepts II — both reversed.
Key takeaway. Erase, don't measure: the eigenvalue register must be uncomputed for the components to interfere into $|x\rangle$.

Q.50 Assertion–Reason · Choice of C

Assertion (A): In HHL the constant C must satisfy $C \leq \min_j |\lambda_j|$ (more precisely, $\leq$ the smallest $|\tilde{\lambda}|$ that can occur).

Reason (R): The controlled rotation puts the amplitude $C/\tilde{\lambda}$ on the ancilla state $|1\rangle$, and an amplitude cannot exceed 1 in magnitude.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. The controlled rotation needs $|\sin\frac{\theta}{2}| = |C/\tilde{\lambda}| \leq 1$; C sets the overall scale of the success amplitude.

Relevant formulae. $\sin\frac{\theta}{2} = \frac{C}{\tilde{\lambda}}$, $|C| \leq \min|\tilde{\lambda}|$; $P_{\text{success}} \propto C^2$.

Step-by-step explanation. **R is true:** $R_Y(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$ with $\sin\frac{\theta}{2} = C/\tilde{\lambda}$, which requires $|C/\tilde{\lambda}| \leq 1$. **A is true** and follows from R: for every eigenvalue that occurs, $|C/\tilde{\lambda}| \leq 1 \iff C \leq |\tilde{\lambda}|$, so C is bounded by the smallest one. (The notebook sets $C = 2\pi/(2^{n_c}t)$, the smallest non-zero representable value, and clips the argument of arcsin to $[-1, 1]$.) Choosing C as large as allowed maximises the success probability $\propto C^2$. Answer: (A).

Why the other options are incorrect.

(B) denies the link, but the bound on C is exactly the requirement that the amplitude $C/\tilde{\lambda}$ be a valid sine.

(C) rejects R, but R_Y puts $\sin\frac{\theta}{2}$ on $|1\rangle$, which is bounded by 1.

(D) rejects A; a larger C would make $\arcsin(C/\tilde{\lambda})$ undefined for the smallest eigenvalue.

Key takeaway. Choose C as large as allowed, i.e. about $\lambda_{\min}$; then $P_{\text{success}} \gtrsim 1/\kappa^2$.

Q.51 Single correct · Negative eigenvalues (two's complement)

In the default notebook circuit ($n_c = 4$, $t = \pi/8$, $C = 1$, two's-complement clock), the clock value $k = 14$ (1110) is treated as the eigenvalue $\tilde{\lambda}$ and rotated by θ, where

(A) $\tilde{\lambda} = -2$, $\theta = +\pi/3$

(B) $\tilde{\lambda} = 2$, $\theta = \pi/3$

(C) ✓ $\tilde{\lambda} = -2$, $\theta = -\pi/3$

(D) $\tilde{\lambda} = 14$, $\theta = 2\arcsin\frac{1}{14}$

Answer: (C) Detailed solution

Correct answer. **(C)** $\tilde{\lambda} = -2$, $\theta = -\pi/3$

Concepts required. Negative eigenvalues produce phases $e^{i\lambda t}$ with $\lambda t < 0$, which wrap around to clock values near 2^{n_c} (two's complement).

Relevant formulae. signed $k = k - 2^{n_c}$ for $k \geq 2^{n_c-1}$; $\tilde{\lambda} = kC$; $\theta = 2\arcsin(C/\tilde{\lambda})$, an odd function.

Step-by-step explanation. $k = 14 \geq 2^{n_c-1} = 8$, so it represents $14 - 16 = -2$ (two's complement), and $\tilde{\lambda} = -2\,C = -2$. (Physically: $e^{-2it} = e^{i(2\pi-2t)}$, so a negative eigenvalue produces the phase of $k = 16 - 2 = 14$.) $\theta = 2\arcsin\frac{C}{\tilde{\lambda}} = 2\arcsin(-\frac{1}{2}) = -\frac{\pi}{3}$: the ancilla amplitude is $-\frac{1}{2}$. The *sign* matters — it is the sign of $1/\lambda$ in A^{-1}; $+\pi/3$ would solve the system with $|A|$ instead of A.

Why the other options are incorrect.

(A) $\theta = +\pi/3$ loses the sign: arcsin is odd, and the sign of $1/\lambda$ must survive.

(B) $\tilde{\lambda} = 2$ drops the sign entirely, i.e. inverts $|A|$ instead of A.

(D) $\tilde{\lambda} = 14$ reads the clock as unsigned; with $t = \pi/8$ an eigenvalue 14 would alias, and the notebook interprets $k \geq 8$ as negative.

Key takeaway. Signs of eigenvalues are kept by two's complement and by the sign of the rotation angle.

Q.52 SINGLE CORRECT · Code: X gates of the inversion

The eigenvalue-inversion loop of `HHL.ipynb` runs over $k = 1, \ldots, 2^{n_c} - 1$; for each k it applies `kernel.x` to every clock qubit whose bit of k is 0, then the multi-controlled `cry`, then the same `x` gates again. For $n_c = 4$ the total number of `x` gates is

(A) ✓ 56 **(B)** 28 **(C)** 60 **(D)** 32

Answer: (A) Detailed solution

Correct answer. (A) 56

Concepts required. Each clock value k is selected by flipping its zero bits, applying the multi-controlled rotation, and flipping back.

Relevant formulae. $\#X = 2\sum_{k=1}^{2^{n_c}-1}(n_c - \text{popcount}(k))$; $\sum_{k=0}^{2^n-1}\text{popcount}(k) = n2^{n-1}$.

Step-by-step explanation. For each k the number of zero bits is $4 - \text{popcount}(k)$, and each zero bit costs 2 gates. Summed over $k = 1, \ldots, 15$: $\sum_k \text{popcount}(k) = 4 \cdot 2^3 = 32$ (each bit is 1 in half of the 16 values; $k = 0$ adds nothing), so

$$2\sum_{k=1}^{15}(4 - \text{popcount}(k)) = 2\,(60 - 32) = 56.$$

This matches the recorded circuit statistics (`'x': 56`, `'ccccry': 15`). 28 forgets the second (restoring) layer.

Why the other options are incorrect.

(B) 28 counts only the first layer of X gates, not the restoring layer.

(C) $60 = 15 \times 4$ assumes every clock value has four zero bits.

(D) 32 counts the one-bits (popcount sum) instead of the zero bits.

Key takeaway. Code-level counts can be checked against `cudaq.estimate_resources`: the recorded statistics show `'x': 56`.

Q.53 SINGLE CORRECT · Norm recovery $\vec{x} = \|\vec{b}\|\,\vec{a}/C$

Let $\vec{a}$ be the (unnormalised) amplitudes of the system register in the branch clock $= 0\ldots0$, ancilla $= 1$ of an exact HHL run. The notebook recovers the solution, including its norm, as

(A) $\vec{x} = \dfrac{\vec{a}}{\sqrt{P_{\text{success}}}}$ **(B)** ✓ $\vec{x} = \dfrac{\|\vec{b}\|}{C}\,\vec{a}$

(C) $\vec{x} = \dfrac{C}{\|\vec{b}\|}\,\vec{a}$ **(D)** $\vec{x} = \vec{a}$

Answer: (B) Detailed solution

Correct answer. (B) $\vec{x} = \dfrac{\|\vec{b}\|}{C}\,\vec{a}$

Concepts required. The post-selected branch is $C\,A^{-1}\,|b\rangle$; $|b\rangle$ is $\vec{b}$ divided by its norm.

Relevant formulae. $\vec{a} = \frac{C}{\|\vec{b}\|}\vec{x}$, $\|\vec{a}\|^2 = P_{\text{success}}$, so $\vec{x} = \frac{\|\vec{b}\|}{C}\vec{a}$.

Step-by-step explanation. The surviving branch is $\vec{a} = \sum_j \beta_j \frac{C}{\lambda_j}\,|u_j\rangle = C\,A^{-1}\,|b\rangle = C\,A^{-1}\frac{\vec{b}}{\|\vec{b}\|} =$

$\frac{C}{\|\vec{b}\|}\vec{x}$. Solving for $\vec{x}$: $\vec{x} = \frac{\|\vec{b}\|}{C}\vec{a}$ (code: `x_quantum_full = amplitudes * b_norm / C_constant`). $\vec{a}/\sqrt{P_{\text{success}}}$ is the *normalised* $|x\rangle$ (correct direction, norm 1), not $\vec{x}$. Note that $\|\vec{a}\|^2 = P_{\text{success}}$, so on hardware the norm of $\vec{x}$ can be estimated from the measured success rate.

Why the other options are incorrect.

(A) $\vec{a}/\sqrt{P_{\text{success}}}$ is the normalised $|x\rangle$ — correct direction, but norm 1.

(C) $C/\|\vec{b}\|$ inverts the conversion factor.

(D) $\vec{a}$ itself equals $\vec{x}$ only if $\|\vec{b}\| = C$ (true in the default run, but not in general).

Key takeaway. Scale information lives in $\|\vec{b}\|$, C and P_{success}, not in the normalised state.

Q.54 SINGLE CORRECT · Default 4×4 solution

For the notebook's default system (A with eigenvalues $1, 2, 3, 4$ and $\vec{b} = (1, 0, 0, 0)$, see conventions) the exact solution is

(A) ✓ $\vec{x} = \frac{1}{48}(25,\ 7,\ 11,\ 5)$

(B) $\vec{x} = \frac{1}{48}(25,\ -7,\ -11,\ 5)$

(C) $\vec{x} = \frac{1}{\sqrt{820}}(25,\ 7,\ 11,\ 5)$

(D) $\vec{x} = (1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4})$

Answer: (A) **Detailed solution**

Correct answer. **(A)** $\vec{x} = \frac{1}{48}(25,\ 7,\ 11,\ 5)$

Concepts required. The default matrix is $H_4\,\mathrm{diag}(1,2,3,4)H_4$ with the self-inverse orthogonal H_4; invert in the eigenbasis.

Relevant formulae. $A^{-1}\vec{b} = H_4\Lambda^{-1}H_4\vec{b}$; $H_4\vec{b} = \frac{1}{2}(1,1,1,1)$ for $\vec{b} = (1,0,0,0)$.

Step-by-step explanation. $A = H_4\Lambda H_4$ with $H_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ ($H_4^2 = I$), so $A^{-1} = H_4\Lambda^{-1}H_4$. $H_4\vec{b} = \frac{1}{2}(1,1,1,1)$ (all $\beta_j = \frac{1}{2}$); Λ^{-1} gives $\frac{1}{2}(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}) = \frac{1}{24}(12, 6, 4, 3)$; then

$$H_4 \cdot \tfrac{1}{24}(12,6,4,3) = \tfrac{1}{48}(12{+}6{+}4{+}3,\ 12{-}6{+}4{-}3,\ 12{+}6{-}4{-}3,\ 12{-}6{-}4{+}3) = \tfrac{1}{48}(25,7,11,5).$$

$= (0.520833, 0.145833, 0.229167, 0.104167)$, exactly the recorded output. Check the first row: $2.5 \cdot 25 - 0.5 \cdot 7 - 1 \cdot 11 = 48$. $\frac{1}{\sqrt{820}}(\dots)$ is the normalised $|x\rangle$; $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4})$ inverts the eigenvalues without changing basis.

Why the other options are incorrect.

(B) Negative signs would require negative entries of $A^{-1}\vec{b}$; all four sums $12 \pm 6 \pm 4 \pm 3$ are positive.

(C) $\frac{1}{\sqrt{820}}(25, 7, 11, 5)$ is the normalised state $|x\rangle$, not the solution vector ($\|\vec{x}\| \approx 0.597$).

(D) $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4})$ inverts the eigenvalues but forgets to transform to and from the eigenbasis.

Key takeaway. Recorded result $= \frac{1}{48}(25, 7, 11, 5)$ exactly; fidelity 1 because every eigenvalue hits an integer clock value.

Q.55 MATCHING LISTS · Algorithm step ↔ CUDA-Q code

Match each step of HHL (List-I) with the CUDA-Q / Python code from `HHL.ipynb` that implements it (List-II).

List-I		List-II	
(P)	amplitude encoding of $\|b\rangle$	**(1)**	`kernel.cry(theta, clock, ancilla)`
(Q)	controlled $U^{2^j} = e^{iAt2^j}$	**(2)**	`cudaq.register_operation(name, matrix_power(U, 2**j))`
(R)	eigenvalue inversion	**(3)**	`bits[:n_c] == '0'*n_c and bits[-1] == '1'`
(S)	post-selection of the shots	**(4)**	`kernel.qalloc(initial_state_vector)`

(A) P→4, Q→1, R→2, S→3
(B) P→4, Q→2, R→3, S→1
(C) P→2, Q→4, R→1, S→3
(D) ✓ P→4, Q→2, R→1, S→3

Answer: (D) Detailed solution

Correct answer. **(D)** P→4, Q→2, R→1, S→3
Concepts required. Mapping the algorithm onto the notebook's kernel-builder code.
Relevant formulae. Bit strings are ordered clock | system | ancilla; success: clock $= 0\ldots0$, ancilla $= 1$.
Step-by-step explanation.

- P: the initial state $|0\ldots0\rangle_{\text{clock}}|b\rangle|0\rangle$ is loaded in one step with `qalloc(vector)` (4).
- Q: each U^{2^j} is an exact matrix registered as a custom gate and applied controlled by one clock qubit (2).
- R: the multi-controlled $R_Y(\theta_k)$ on the ancilla, $\theta_k = 2\arcsin(C/\tilde{\lambda}_k)$ (1).
- S: in the sampled bit strings (clock | system | ancilla) keep clock $= 0\ldots0$ and ancilla $= 1$ (3).

Why the other options are incorrect.
(A) Q→1, R→2: the `cry` is the rotation, the registered matrix powers are the controlled evolutions.
(B) R→3, S→1: post-selection is classical filtering of bit strings, the rotation is a gate.
(C) P→2, Q→4 swaps state preparation (`qalloc`) with the registered matrix powers.
Key takeaway. Four lines of code carry the four ideas of HHL: load $|b\rangle$, controlled e^{iAt2^j}, controlled R_Y, post-select.

Q.56 SINGLE CORRECT · Preparing $|b\rangle$

Starting from $|00\rangle$, which circuit prepares $|b\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$, i.e. $\vec{b} \propto (1,-1,1,-1)$?

(A) H on both qubits only
(B) ✓ H on both qubits, then Z on q_1
(C) H on both qubits, then Z on q_0
(D) X on both qubits, then H on both qubits

Answer: (B) Detailed solution

Correct answer. **(B)** H on both qubits, then Z on q_1
Concepts required. Product vectors are prepared with single-qubit gates; signs come from Z (or from H acting on $|1\rangle$).
Relevant formulae. $|+\rangle|-\rangle = \frac{1}{2}(1,-1,1,-1)$; $Z|+\rangle = |-\rangle$; $H|1\rangle = |-\rangle$.
Step-by-step explanation. Factorise: $\frac{1}{2}(|0\rangle + |1\rangle)_{q_0}(|0\rangle - |1\rangle)_{q_1} = |+\rangle|-\rangle$. $H|0\rangle = |+\rangle$ on q_0; on q_1, $ZH|0\rangle = Z|+\rangle = |-\rangle$.

- Z on q_0 gives $|-\rangle|+\rangle \propto (1,1,-1,-1)$; $H\otimes H$ alone gives $(1,1,1,1)$; $X\otimes X$ then $H\otimes H$ gives $|-\rangle|-\rangle \propto (1,-1,-1,1)$.

Structured vectors like this are cheap to prepare; a *general* $\vec{b}$ needs $O(d)$ gates (or a QRAM), which can cancel HHL's speed-up — the notebook side-steps this with exact amplitude initialisation on a simulator.

Why the other options are incorrect.

(A) $H\otimes H$ alone gives $(1,1,1,1)/2$ — no minus signs.

(C) Z on q_0 produces $|-\rangle|+\rangle \propto (1,1,-1,-1)$.

(D) $X\otimes X$ then $H\otimes H$ gives $|-\rangle|-\rangle \propto (1,-1,-1,1)$.

Key takeaway. Efficient preparation of $|b\rangle$ is an assumption of HHL; structured vectors satisfy it, arbitrary data may not.

Q.57 SINGLE CORRECT · Role of the inverse QFT

Inside the phase-estimation part of HHL, the inverse QFT on the clock register

(A) computes $1/\lambda$ for every eigenvalue

(B) ✓ turns the kicked-back phases $e^{2\pi i\, kj/2^{n_c}}$ into the computational-basis state $|k\rangle$ (the binary eigenvalue estimate)

(C) measures the eigenvalues and sends them to the classical computer

(D) prepares $|b\rangle$ in the eigenbasis of A

Answer: (B) Detailed solution

Correct answer. **(B)** turns the kicked-back phases $e^{2\pi i\, kj/2^{n_c}}$ into the computational-basis state $|k\rangle$ (the binary eigenvalue estimate)

Concepts required. QPE = Hadamards + controlled powers (phase kickback creates a Fourier state) + inverse QFT (reads it out in binary).

Relevant formulae. $\frac{1}{\sqrt{2^{n_c}}}\sum_j e^{2\pi ijk/2^{n_c}}|j\rangle = \mathrm{QFT}\,|k\rangle$; QFT^{-1} gives $|k\rangle$.

Step-by-step explanation. After the Hadamards and controlled powers the clock holds $\frac{1}{\sqrt{2^{n_c}}}\sum_j e^{i\lambda tj}|j\rangle = \mathrm{QFT}\,|k\rangle$ with $k = \lambda t 2^{n_c}/2\pi$ (phase kickback). The inverse QFT maps this Fourier-basis state back to $|k\rangle$, so the eigenvalue is written as a binary number that a controlled rotation can read. The division by λ happens later (rotation), nothing is measured, and the eigenbasis expansion of $|b\rangle$ is never computed explicitly — linearity does it.

Why the other options are incorrect.

(A) The inverse QFT computes no reciprocals; $1/\lambda$ comes from the controlled rotation.

(C) Nothing is measured in the middle of HHL; the clock stays coherent and is later uncomputed.

(D) $|b\rangle$ is prepared before QPE, and the eigenbasis expansion is never performed explicitly.

Key takeaway. Inverse QFT: Fourier-encoded phase → binary eigenvalue register.

Q.58 Statement I/II · $|b\rangle$ an eigenvector

Assume $|b\rangle$ is itself an eigenvector $|u_j\rangle$ of A and λ_j is exactly representable by the clock.

Statement I: The success probability is C^2/λ_j^2.

Statement II: After phase estimation the clock register is in the single basis state $|k_j\rangle$ (not entangled with the system).

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true

Concepts required. With a single eigencomponent there is nothing to interfere: QPE gives a product state and HHL reduces to one rotation.

Relevant formulae. $\beta_j = 1 \Rightarrow P_{\text{success}} = C^2/\lambda_j^2$; $|0\rangle|u_j\rangle \to |k_j\rangle|u_j\rangle$.

Step-by-step explanation. With $\beta_j = 1$ and all other β zero: **I is true:** $P = \sum|\beta|^2 C^2/\lambda^2 = C^2/\lambda_j^2$. **II is true:** QPE maps $|0\rangle|u_j\rangle \to |k_j\rangle|u_j\rangle$, a product state; with an exactly representable phase the clock is a single basis state. The output is $|x\rangle = |u_j\rangle$ (since $A^{-1}|u_j\rangle = \lambda_j^{-1}|u_j\rangle$), as expected. Answer: (A).

Why the other options are incorrect.

(B) "both false" contradicts the general formulas specialised to one component.

(C) rejects II, but an exact single phase gives a single clock basis state.

(D) rejects I, but $\sum_j |\beta_j|^2 C^2/\lambda_j^2$ has only one term.

Key takeaway. Eigenvector input is the simplest sanity test of an HHL implementation.

Paragraph for Questions 59 and 60

HHL is run by hand on $A = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$, $\vec{b} = (1,0)$ $(n_b = 1)$. The eigenpairs are $\lambda_1 = 1$, $|u_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and $\lambda_2 = 2$, $|u_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Phase estimation uses $n_c = 2$ clock qubits and $t = \pi/2$; because both eigenvalues are positive, the clock value k is read as an *unsigned* integer here $(\tilde{\lambda}_k = kC)$. States are written as $|\text{clock}\rangle|\text{system}\rangle$; the ancilla starts in $|0\rangle$ and is not shown until the rotation.

Q.59 Paragraph based · 2×2 HHL trace: after QPE

The state after phase estimation is

(A) $\frac{1}{\sqrt{2}}(|00\rangle|u_1\rangle + |01\rangle|u_2\rangle)$

(B) $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \otimes |0\rangle$

(C) $\frac{1}{\sqrt{2}}(|10\rangle|u_1\rangle + |01\rangle|u_2\rangle)$

(D) ✓ $\frac{1}{\sqrt{2}}(|01\rangle|u_1\rangle + |10\rangle|u_2\rangle)$

Answer: (D) Detailed solution

Correct answer. **(D)** $\frac{1}{\sqrt{2}}(|01\rangle|u_1\rangle + |10\rangle|u_2\rangle)$

Concepts required. Expand $|b\rangle$ in the eigenbasis and apply QPE to each component separately.

Relevant formulae. $|0\rangle = \frac{1}{\sqrt{2}}(|u_1\rangle + |u_2\rangle)$; $k = \lambda t 2^{n_c}/2\pi = \lambda$ for $t = \pi/2$, $n_c = 2$.

Step-by-step explanation. Expand $|b\rangle = |0\rangle = \frac{1}{\sqrt{2}}(|u_1\rangle + |u_2\rangle)$, so $\beta_1 = \beta_2 = \frac{1}{\sqrt{2}}$. Clock values: $k = \lambda t 2^{n_c}/2\pi = \lambda \cdot \frac{\pi}{2} \cdot \frac{4}{2\pi} = \lambda$, so $\lambda_1 = 1 \to |01\rangle$ and $\lambda_2 = 2 \to |10\rangle$, both exact. Hence $\frac{1}{\sqrt{2}}(|01\rangle|u_1\rangle + |10\rangle|u_2\rangle)$: clock and system are entangled. Also $C = 2\pi/(2^{n_c}t) = 2\pi/(4 \cdot \frac{\pi}{2}) =$

1. (Option with $|0\rangle$ factored out ignores this entanglement; the swapped option exchanges the eigenvalues.)

Why the other options are incorrect.

(A) $|00\rangle$ and $|01\rangle$ correspond to $k = \lambda - 1$; $k = 0$ would mean $\lambda = 0$.

(B) Factoring out the system ignores that each clock value belongs to a different eigenvector (entanglement).

(C) Swapping $|01\rangle$ and $|10\rangle$ assigns $\lambda = 2$ to $|u_1\rangle$ and $\lambda = 1$ to $|u_2\rangle$.

Key takeaway. After QPE each eigenvector carries its own eigenvalue label.

Q.60 PARAGRAPH BASED · 2×2 HHL TRACE: OUTPUT

After the controlled rotation, uncomputation and a successful ancilla measurement, the success probability and the system state are

(A) $P = \frac{5}{8}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle + |1\rangle)$

(B) $P = \frac{3}{4}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$

(C) ✓ $P = \frac{5}{8}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$

(D) $P = 1$; $|0\rangle$

Answer: (C) **Detailed solution**

Correct answer. **(C)** $P = \frac{5}{8}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$

Concepts required. Rotation multiplies each component by C/λ_j; after uncomputation the components interfere into $C\,A^{-1}\,|b\rangle$.

Relevant formulae. branch $= \sum_j \beta_j \frac{C}{\lambda_j}\,|u_j\rangle$; $P = \|\text{branch}\|^2$; $|x\rangle = \text{branch}/\sqrt{P}$.

Step-by-step explanation. Rotation: $|01\rangle \to$ ancilla amplitude $C/1 = 1$; $|10\rangle \to C/2 = \frac{1}{2}$. After uncomputation the ancilla-1 branch is

$$\tfrac{1}{\sqrt{2}}\Big(1 \cdot |u_1\rangle + \tfrac{1}{2}\,|u_2\rangle\Big) = \tfrac{1}{2}\Big[(|0\rangle - |1\rangle) + \tfrac{1}{2}(|0\rangle + |1\rangle)\Big] = \tfrac{3}{4}\,|0\rangle - \tfrac{1}{4}\,|1\rangle\,.$$

$P = (\frac{3}{4})^2 + (\frac{1}{4})^2 = \frac{10}{16} = \frac{5}{8}$ $\left(= \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot\frac{1}{4}\right)$, and the normalised state is $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$. Classical check: $A^{-1}\vec{b} = \frac{1}{2}\begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}(1,0)^T = (\frac{3}{4}, -\frac{1}{4})$ — with $\|\vec{b}\| = C = 1$ the branch *is* $\vec{x}$. ($P = \frac{3}{4}$ forgets to square the amplitudes.)

Why the other options are incorrect.

(A) $+\,|1\rangle$ has the wrong sign: $|u_1\rangle$ (weight 1) contributes $-\,|1\rangle$, $|u_2\rangle$ (weight $\frac{1}{2}$) only $+\frac{1}{2}\,|1\rangle$.

(B) $P = \frac{3}{4}$ adds the amplitudes $\frac{1}{2}(1 + \frac{1}{2})$ instead of the probabilities $\frac{1}{2}(1 + \frac{1}{4})$.

(D) $P = 1$ and $|0\rangle$ would require $A = I$; the run succeeds only with probability $\frac{5}{8}$.

Key takeaway. A full HHL run by hand: $|x\rangle = \frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle) \propto A^{-1}\vec{b} = (\frac{3}{4}, -\frac{1}{4})$.

SECTION D · Advanced HHL and Implementation Q.61–80

Q.61 Single correct · Complexity of HHL

For an s-sparse Hermitian $N \times N$ matrix with condition number κ, target error ϵ, and efficient preparation of $|b\rangle$, the run time of the original HHL algorithm (2009) scales as

(A) $O(\log N)$, independent of κ and ϵ
(B) $\tilde{O}(N\, s\, \kappa)$
(C) ✓ $\tilde{O}(\log(N)\, s^2\kappa^2/\epsilon)$
(D) $O(N^3)$

Answer: (C) Detailed solution

Correct answer. (C) $\tilde{O}(\log(N)\, s^2\kappa^2/\epsilon)$
Concepts required. HHL's cost combines sparse Hamiltonian simulation, phase-estimation precision and the number of repetitions.
Relevant formulae. $T_{\text{HHL}} = \tilde{O}(\log(N)s^2\kappa^2/\epsilon)$; Gaussian elimination $O(N^3)$; conjugate gradient $O(Ns\sqrt{\kappa}\log(1/\epsilon))$.
Step-by-step explanation. Harrow, Hassidim and Lloyd showed a run time $\tilde{O}(\log(N)s^2\kappa^2/\epsilon)$: $\log N$ from the number of qubits and sparse Hamiltonian simulation, s^2 from simulating e^{iAt} for an s-sparse matrix, one factor κ/ϵ from the phase-estimation precision and one factor κ from the $O(\kappa^2)$ repetitions reduced to $O(\kappa)$ by amplitude amplification. $O(N^3)$ is Gaussian elimination; $\tilde{O}(Ns\kappa)$ is roughly classical conjugate gradient ($Ns\sqrt{\kappa}$ for positive-definite A). No algorithm is independent of κ.
Why the other options are incorrect.
(A) No linear-system algorithm is independent of κ: even the optimal quantum algorithm needs $\Omega(\kappa)$.
(B) $\tilde{O}(Ns\kappa)$ is the classical iterative (conjugate-gradient-type) scaling, linear in N.
(D) $O(N^3)$ is classical Gaussian elimination.
Key takeaway. Exponential improvement in N, polynomial cost in κ and $1/\epsilon$ — the speed-up survives only if κ is small.

Q.62 Single correct · Repetitions and amplitude amplification

Suppose $P_{\text{success}} = 1/\kappa^2$ with $\kappa = 10$. The expected number of repetitions needed for one successful run without, and with, amplitude amplification is about

(A) 1 without; 1 with
(B) ✓ 100 without; 10 with
(C) 100 without; 100 with (no improvement is possible)
(D) 10 without; $\sqrt{10}$ with

Answer: (B) Detailed solution

Correct answer. (B) 100 without; 10 with
Concepts required. Repeat-until-success needs $1/P$ trials on average; amplitude amplification needs only $O(1/\sqrt{P})$.
Relevant formulae. $\mathbb{E}[\text{trials}] = 1/P$; amplitude amplification: $\approx \frac{\pi}{4\sqrt{P}}$ iterations.
Step-by-step explanation. Repeat-until-success: the number of trials is geometric with mean $1/P = \kappa^2 = 100$. Amplitude amplification (Grover-type rotations around the success subspace) raises the success amplitude $\sqrt{P} = 1/\kappa$ to $O(1)$ with $O(1/\sqrt{P}) = O(\kappa) \approx 10$ applications of the HHL circuit and its inverse — a quadratic saving. This is how the original analysis turns κ^2 repetitions into a factor κ in the run time. (It requires coherent repetitions, not just re-running and measuring.)

Why the other options are incorrect.

(A) $P = 0.01$ is far from 1: most runs fail.

(C) Amplitude amplification gives a quadratic improvement, so $100 \to 10$ is possible.

(D) 10 without amplification would be $1/\sqrt{P}$; repetitions scale like $1/P$.

Key takeaway. Post-selection costs κ^2 repetitions; amplitude amplification brings it down to κ.

Q.63 MATCHING LISTS · Quantum linear-system algorithms

Match each quantum linear-system algorithm (List-I) with its characteristic improvement or complexity (List-II).

List-I		List-II	
(P)	HHL (Harrow, Hassidim, Lloyd, 2009)	**(1)**	precision dependence polylog$(1/\epsilon)$ via Fourier/Chebyshev fitting of $1/x$ (LCU)
(Q)	Variable-time amplitude amplification (Ambainis, 2012)	**(2)**	almost linear dependence on κ
(R)	Childs, Kothari and Somma (2017)	**(3)**	$\tilde{O}(\kappa^2/\epsilon)$ dependence on κ and ϵ
(S)	Discrete adiabatic method (Costa et al., 2022)	**(4)**	optimal $O(\kappa \log(1/\epsilon))$ query complexity

(A) P→2, Q→3, R→1, S→4

(B) P→3, Q→2, R→4, S→1

(C) ✓ P→3, Q→2, R→1, S→4

(D) P→3, Q→1, R→2, S→4

Answer: (C) Detailed solution

Correct answer. **(C)** P→3, Q→2, R→1, S→4

Concepts required. Quantum linear-system algorithms improved HHL's κ and ϵ dependence step by step.

Relevant formulae. HHL $\tilde{O}(\kappa^2/\epsilon)$; VTAA $\tilde{O}(\kappa)$; CKS polylog$(1/\epsilon)$; discrete adiabatic $O(\kappa \log \frac{1}{\epsilon})$.

Step-by-step explanation.

- P: HHL uses phase estimation, which gives κ^2/ϵ (3).
- Q: Ambainis' variable-time amplitude amplification stops "branches" with large eigenvalues early and reaches $\tilde{O}(\kappa \log^3 \kappa)$ (2).
- R: Childs–Kothari–Somma replace phase estimation by a linear combination of unitaries approximating $1/x$, giving polylog$(1/\epsilon)$ (1).
- S: Costa et al. use a discrete adiabatic walk and reach the optimal $O(\kappa \log(1/\epsilon))$ (4). Quantum singular value transformation (QSVT) gives a unified framework for these methods.

Why the other options are incorrect.

(A) P→2, Q→3 attributes the original κ^2/ϵ to Ambainis and near-linear κ to HHL.

(B) R→4, S→1: the optimal $O(\kappa \log(1/\epsilon))$ belongs to the discrete adiabatic method, not to CKS.

(D) Q→1, R→2 swaps Ambainis (improved κ) and Childs–Kothari–Somma (improved ϵ).

Key takeaway. Modern methods reach optimal $\kappa \log(1/\epsilon)$; the read-out and input caveats remain.

Q.64 SINGLE CORRECT · Read-out cost of all x_i

Why can HHL *not* efficiently output all d components of $\vec{x}$ for $d = 2^{20}$?

(A) The no-cloning theorem forbids measuring the system register

(B) ✓ Sampling gives each $|x_i|^2$ only statistically; estimating all d values needs $\Omega(d)$ repetitions (about d/ϵ^2 for precision ϵ), destroying the $\log d$ scaling

(C) Complex amplitudes cannot be measured at all

(D) The ancilla measurement destroys $|x\rangle$ every time, so the system register never contains $|x\rangle$

Answer: (B) Detailed solution

Correct answer. **(B)** Sampling gives each $|x_i|^2$ only statistically; estimating all d values needs $\Omega(d)$ repetitions (about d/ϵ^2 for precision ϵ), destroying the $\log d$ scaling
Concepts required. Sampling a d-dimensional distribution to precision ϵ needs about d/ϵ^2 samples.
Relevant formulae. one run $\to$ one sample i with $P(i) = |x_i|^2/\|\vec{x}\|^2$; tomography $\sim d/\epsilon^2$ runs.
Step-by-step explanation. One run yields one sample i with probability $|x_i|^2/\|\vec{x}\|^2$. To see all d components even once needs at least $\sim d$ samples, and to estimate each with additive precision ϵ needs about d/ϵ^2 (tomography); signs and phases need further measurement bases. Each sample costs a full HHL run, so the total is $\Omega(d)$ — no better than classical. Measurement itself is allowed (no-cloning only forbids *copying* unknown states); a successful ancilla outcome *leaves* $|x\rangle$ in the system register.
Why the other options are incorrect.

(A) No-cloning forbids *copying* an unknown state; measuring it is allowed.

(C) Complex amplitudes are inferred from measurements in several bases — possible, just expensive.

(D) A successful ancilla outcome leaves $|x\rangle$ in the system register; only failed runs lose it.

Key takeaway. Printing $\vec{x}$ costs $\Omega(d)$ — use HHL for global properties of $\vec{x}$.

Q.65 ASSERTION–REASON · "Exponential speed-up" claims

Assertion (A): HHL gives an exponential speed-up for solving any linear system $A\vec{x} = \vec{b}$ and printing the solution $\vec{x}$.

Reason (R): Under suitable assumptions (sparse, well-conditioned A, efficient preparation of $|b\rangle$, only a property of $|x\rangle$ wanted) the run time of HHL is polylogarithmic in the dimension.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) ✓ A is false, but R is true

Answer: (D) Detailed solution

Correct answer. **(D)** A is false, but R is true
Concepts required. HHL's polylogarithmic run time holds only under several assumptions; printing $\vec{x}$ violates one of them.
Relevant formulae. $\tilde{O}(\log(N)s^2\kappa^2/\epsilon)$ assumes sparse A, small κ, efficient $|b\rangle$, and output = a property of $|x\rangle$.
Step-by-step explanation. **A is false**, for three independent reasons: "any" system ignores the sparsity and κ requirements (for $\kappa \sim d$ the advantage vanishes), loading an arbitrary $\vec{b}$ can cost $O(d)$, and printing $\vec{x}$ needs $\Omega(d)$ repetitions (Q.64). **R is true**: this is precisely the HHL result, $\tilde{O}(\log(N)s^2\kappa^2/\epsilon)$. Aaronson's "read the fine print" lists exactly these caveats. Answer: (D).
Why the other options are incorrect.

(A) needs A true, but A overstates HHL: not "any" system (sparsity, κ) and not printing $\vec{x}$ ($\Omega(d)$ read-out).

(B) also needs A true; the fine print makes A false.

(C) needs A true and R false, but A is false and R is precisely the HHL theorem.

Key takeaway. Read the fine print: sparse, well conditioned, cheap $|b\rangle$, and only an expectation value as output.

Q.66 SINGLE CORRECT · Too few clock qubits: rescaled t

The notebook is run with `NUM_CLOCK_QUBITS = 3` on the default matrix (eigenvalues $1, 2, 3, 4$). Since $\kappa = 4$ exceeds the $2^{n_c-1} - 1 = 3$ positive levels, "auto" sets $t = 2\pi \cdot 3/(2^{n_c}\lambda_{\max})$. The ideal clock values $k_j = \lambda_j t 2^{n_c}/2\pi$ are

(A) ✓ 0.75, 1.5, 2.25, 3: only $\lambda = 4$ is exactly representable

(B) 1, 2, 3, 4: all exact

(C) 0.5, 1, 1.5, 2: $\lambda = 2$ and 4 exact

(D) 1.5, 3, 4.5, 6: aliasing of $\lambda = 3, 4$

Answer: (A) Detailed solution

Correct answer. **(A)** 0.75, 1.5, 2.25, 3: only $\lambda = 4$ is exactly representable

Concepts required. If κ exceeds the number of positive clock levels, t must be reduced so the largest eigenvalue fits; small eigenvalues then become non-integer.

Relevant formulae. $t = \frac{2\pi k_{\max}}{2^{n_c}\lambda_{\max}}$, $k_{\max} = 2^{n_c-1} - 1$; $k_j = \lambda_j t 2^{n_c}/2\pi$; $C = 2\pi/(2^{n_c}t)$.

Step-by-step explanation. $t = \frac{2\pi\cdot 3}{8\cdot 4} = \frac{3\pi}{16}$. Then $k_j = \lambda_j \cdot \frac{3\pi}{16} \cdot \frac{8}{2\pi} = \frac{3}{4}\lambda_j$: $0.75, 1.5, 2.25, 3$. Only 3 is an integer, so phase estimation rounds the other three (with spread-out probability, Q.29) and the eigenvalue inversion uses wrong values of $1/\lambda$: the fidelity drops below 1. Also $C = 2\pi/(8t) = \frac{4}{3} > \lambda_{\min} = 1$: the eigenvalue 1 is read either as clock value 1 ($\tilde{\lambda} = \frac{4}{3}$, wrong $1/\lambda$) or as 0 (no rotation, component lost). $0.5, 1, 1.5, 2$ would be $t = \pi/8$ unchanged. This is the "Things to try next" experiment of the notebook.

Why the other options are incorrect.

(B) $1, 2, 3, 4$ would need $k_{\max} \geq 4$, i.e. at least 4 clock qubits.

(C) $0.5, 1, 1.5, 2$ keeps $t = \pi/8$, but "auto" recomputes t when $\kappa > k_{\max}$.

(D) $1.5, 3, 4.5, 6$ would exceed the representable range; "auto" is designed to prevent aliasing.

Key takeaway. Too few clock qubits ⇒ rounded eigenvalues ⇒ wrong $1/\lambda$ ⇒ fidelity < 1.

Q.67 SINGLE CORRECT · Clock qubits for precision ϵ

Using the rule of thumb $n_c \approx \lceil \log_2(\kappa/\epsilon) \rceil$ for the number of clock qubits, a system with $\kappa = 100$ and target relative precision $\epsilon = 0.01$ needs about

(A) 7 clock qubits

(B) 10 clock qubits

(C) 100 clock qubits

(D) ✓ 14 clock qubits

Answer: (D) Detailed solution

Correct answer. **(D)** 14 clock qubits

Concepts required. The clock must resolve the smallest eigenvalue to relative precision ϵ while spanning the largest.

Relevant formulae. $2^{n_c} \gtrsim \kappa/\epsilon \Rightarrow n_c \approx \lceil \log_2(\kappa/\epsilon) \rceil$; $\log_2 10 \approx 3.32$.

Step-by-step explanation. $\kappa/\epsilon = 100/0.01 = 10^4$, $\log_2 10^4 = 4 \log_2 10 \approx 4 \times 3.32 = 13.29$, rounded up: 14. Reason: the smallest eigenvalue $\lambda_{\min} = \lambda_{\max}/\kappa$ must be resolved with relative error

ϵ, i.e. absolute resolution $\epsilon\lambda_{\min}$, while the clock must also span $\lambda_{\max}$: $2^{n_c} \gtrsim \lambda_{\max}/(\epsilon\lambda_{\min}) = \kappa/\epsilon$. ($7 = \lceil \log_2 100 \rceil$ ignores ϵ.) The phase-estimation cost (total evolution time $\propto 2^{n_c}$) grows like κ/ϵ.

Why the other options are incorrect.

(A) $7 = \lceil \log_2 100 \rceil$ uses only κ and ignores the precision.

(B) 10 has no justification from the rule; $2^{10} = 1024 < 10^4$.

(C) 100 confuses the number of qubits with κ; qubits grow logarithmically.

Key takeaway. Clock qubits $\sim \log(\kappa/\epsilon)$, but QPE time $\sim \kappa/\epsilon$: precision is cheap in qubits, expensive in depth.

Q.68 SINGLE CORRECT · Memory of a large HHL simulation

The notebook's state-vector approach is used for a Hermitian $2^{20} \times 2^{20}$ matrix with $n_c = 10$ (fp64, 16 bytes per amplitude). The number of qubits and the memory of the state vector are

(A) 31 qubits; ≈ 17.2 GB

(B) 31 qubits; 496 bytes

(C) $2^{20} + 11$ qubits; beyond any computer

(D) ✓ 31 qubits; ≈ 34.4 GB

Answer: (D) Detailed solution

Correct answer. **(D)** 31 qubits; ≈ 34.4 GB

Concepts required. State-vector memory is $16 \times 2^{\text{qubits}}$ bytes; the exact-matrix approach also stores dense $d \times d$ gates.

Relevant formulae. qubits $= n_b + n_c + 1$; memory $= 16 \cdot 2^n$ B; 2^{30} B ≈ 1.07 GB.

Step-by-step explanation. $n_b = 20$, $n_c = 10$, ancilla 1: 31 qubits. Memory $16 \times 2^{31} = 2^{35}$ bytes $= 34.36 \times 10^9$ bytes ≈ 34.4 GB — more than most single GPUs, i.e. a multi-GPU run. (17.2 GB is 2^{30} amplitudes, one qubit fewer.) Note also that registering e^{iAt2^j} as a dense $2^{20} \times 2^{20}$ custom gate would need 16×2^{40} bytes ≈ 17.6 TB per matrix: exact matrix exponentials do not scale; real implementations use Hamiltonian simulation.

Why the other options are incorrect.

(A) 17.2 GB is 16×2^{30}, i.e. 30 qubits; 31 qubits double it.

(B) 496 bytes is 16×31: one number per qubit instead of per amplitude.

(C) $2^{20} + 11$ counts one qubit per dimension; $n_b = \log_2 2^{20} = 20$.

Key takeaway. Simulating HHL at scale is limited by exponential memory — and by the dense e^{iAt} matrices.

Q.69 SINGLE CORRECT · Trotter steps for e^{iAt}

e^{iAt} with $A = A_1 + A_2$ ($[A_1, A_2] \neq 0$) is approximated by r first-order Trotter steps $\left(e^{iA_1 t/r} e^{iA_2 t/r}\right)^r$. To keep the total error below ϵ, r must grow as

(A) ✓ $r = O(t^2 \|[A_1, A_2]\|/\epsilon)$

(B) $r = O(\log(1/\epsilon))$, independent of t

(C) $r = O(t/\epsilon^2)$

(D) $r = 1$ always suffices, because the error cancels

Answer: (A) Detailed solution

Correct answer. **(A)** $r = O(t^2 \|[A_1, A_2]\|/\epsilon)$

Concepts required. First-order Trotterisation error comes from the commutator and accumulates linearly over the steps.

Relevant formulae. $\|e^{i(A_1+A_2)t} - (e^{iA_1 t/r} e^{iA_2 t/r})^r\| \leq \frac{t^2}{2r} \|[A_1, A_2]\|$.

Step-by-step explanation. Baker–Campbell–Hausdorff: $e^{iA_1\tau} e^{iA_2\tau} = e^{i(A_1+A_2)\tau - \frac{\tau^2}{2}[A_1,A_2] + O(\tau^3)}$,

so each step with $\tau = t/r$ has error $\approx \frac{t^2}{2r^2}\|[A_1, A_2]\|$. Over r steps the errors add: $\approx \frac{t^2}{2r}\|[A_1, A_2]\| \leq \epsilon \Rightarrow r = O(t^2\|[A_1, A_2]\|/\epsilon)$. Higher-order product formulas, qubitisation or QSVT reduce this cost. (The notebook avoids the issue entirely with exact e^{iAt} gates.)

Why the other options are incorrect.

(B) $O(\log(1/\epsilon))$ is achievable only with advanced methods (qubitisation/QSVT), not first-order Trotter, and never independent of t.

(C) $O(t/\epsilon^2)$ has the wrong powers; the error per step is quadratic in t/r.

(D) A single step leaves error $\frac{t^2}{2}\|[A_1, A_2]\|$, which does not cancel for non-commuting terms.

Key takeaway. Hamiltonian simulation is a major cost of real HHL circuits; the notebook replaces it by exact matrices.

Q.70 STATEMENT I/II · What the notebook implements

About `HHL.ipynb`:

Statement I: The controlled evolutions e^{iAt2^j} are implemented as exact matrix exponentials (`scipy.linalg.expm`) registered as custom gates, not by Hamiltonian simulation.

Statement II: The notebook executes HHL on quantum-hardware targets when `BACKEND = "hardware"`.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. (C) Statement I is true, but Statement II is false

Concepts required. The notebook is a simulator demonstration: exact e^{iAt} gates, amplitude encoding, CPU/GPU simulators only.

Relevant formulae. `U_step = expm(1j * A_work * t_evolution)`; allowed methods: `statevector`, `density_matrix`.

Step-by-step explanation. **I is true:** `U_step = expm(1j * A_work * t_evolution)`; each power is registered with `cudaq.register_operation`. **II is false:** the backend cell stops with an error for `"hardware"`: "This HHL implementation uses exact matrix exponentials as custom gates and amplitude encoding, which hardware targets cannot execute. Use a simulator backend." The allowed methods are `statevector` and `density_matrix`; the recorded results come from the CPU simulator `qpp-cpu`. Answer: (C).

Why the other options are incorrect.

(A) accepts II, but the code raises an error for hardware backends.

(B) rejects I, but the controlled evolutions are registered exact matrices.

(D) rejects I and accepts II — both reversed.

Key takeaway. No HHL hardware run was made or is possible with this code; recorded results come from `qpp-cpu`.

Q.71 SINGLE CORRECT · Recorded P_{success}: shot statistics

In the recorded run the ideal success probability is 0.355903 and 7086 of 20 000 shots (0.3543) were accepted. The binomial standard error of the measured fraction and the verdict are

(A) ≈ 0.0007; the deviation is $\approx 2.3\sigma$

(B) ≈ 0.0034; the deviation is $\approx 4.7\sigma$, a circuit error

(C) ✓ ≈ 0.0034; the deviation is $\approx 0.5\sigma$, consistent with shot noise

(D) ≈ 0.012; the deviation is $\approx 0.1\sigma$

Answer: (C) Detailed solution

Correct answer. **(C)** ≈ 0.0034; the deviation is $\approx 0.5\sigma$, consistent with shot noise

Concepts required. The fraction of accepted shots is binomial; compare deviations with its standard error.

Relevant formulae. $\sigma_{\hat{p}} = \sqrt{p(1-p)/N}$; $z = (\hat{p} - p)/\sigma_{\hat{p}}$.

Step-by-step explanation. $\sigma = \sqrt{\frac{p(1-p)}{N}} = \sqrt{\frac{0.355903 \times 0.644097}{20000}} = \sqrt{1.146 \times 10^{-5}} \approx 0.00339$. Deviation: $0.3543 - 0.3559 = -0.0016$, i.e. $-0.0016/0.0034 \approx -0.47\sigma$ — well within statistical fluctuations. ($0.012 \approx 1/\sqrt{7086}$ is the notebook's rough error scale for the *accepted* shots, not this standard error.) The state-vector run gives exactly 0.355903.

Why the other options are incorrect.

(A) 0.0007 is about five times too small: $\sqrt{0.3559 \times 0.6441/20000} = 0.0034$.

(B) 4.7σ uses the wrong deviation (or σ ten times too small); the observed $z \approx -0.5$.

(D) $0.012 = 1/\sqrt{7086}$ is the notebook's error scale for the accepted shots, not for this fraction of 20 000.

Key takeaway. Before blaming the circuit, compare with $\sqrt{p(1-p)/N}$: here everything is consistent.

Q.72 SINGLE CORRECT · Recorded read-out of $|x_0|^2$

In the recorded run, the post-selected shots (7086 accepted) estimate $|x_0|^2/\|\vec{x}\|^2$ as 0.7691; the exact value is 0.7622. The binomial standard error of this estimate and the deviation in units of it are

(A) ✓ ≈ 0.0051; $\approx 1.4\sigma$

(B) ≈ 0.012; $\approx 0.6\sigma$, the exact standard error

(C) ≈ 0.0051; $\approx 14\sigma$

(D) ≈ 0.0030; $\approx 2.3\sigma$

Answer: (A) Detailed solution

Correct answer. **(A)** ≈ 0.0051; $\approx 1.4\sigma$

Concepts required. Read-out statistics use only the accepted (post-selected) shots.

Relevant formulae. $\sigma = \sqrt{p(1-p)/N_{\text{acc}}}$, $N_{\text{acc}} = P_{\text{success}} \cdot \text{SHOTS}$; $\sqrt{p(1-p)} \le \frac{1}{2}$.

Step-by-step explanation. $\sigma = \sqrt{\frac{p(1-p)}{N_{\text{acc}}}} = \sqrt{\frac{0.7622 \times 0.2378}{7086}} = \sqrt{2.558 \times 10^{-5}} \approx 0.00506$. Deviation $0.7691 - 0.7622 = 0.0069 \approx 1.4\sigma$: an ordinary fluctuation. The notebook prints $1/\sqrt{7086} \approx 0.012$ as a conservative scale (since $\sqrt{p(1-p)} \le \frac{1}{2}$, the true standard error is at most 0.0059). Only the accepted shots count: the effective sample size is $P_{\text{success}} \times \text{SHOTS}$, so a low success probability directly increases the read-out error.

Why the other options are incorrect.

(B) $1/\sqrt{N_{\text{acc}}}$ is a conservative scale, not the binomial standard error (which is at most half of it).

(C) 14σ divides 0.0069 by 0.0005; with $\sigma \approx 0.005$ the deviation is 1.4σ.

(D) 0.0030 uses all 20 000 shots; failed shots carry no information about $\vec{x}$.

Key takeaway. Low success probability means fewer useful shots and a larger read-out error.

Q.73 SINGLE CORRECT · Circuit resources vs. n_c

The default circuit ($n_c = 4$, 4×4 matrix) has 15 multi-controlled rotations and 8 controlled $e^{\pm iAt2^j}$ gates on 7 qubits. With the same matrix and $n_c = 5$ the notebook's circuit has

(A) 31 rotations, 10 controlled $e^{\pm iAt2^j}$, 7 qubits

(B) ✓ 31 rotations, 10 controlled $e^{\pm iAt2^j}$, 8 qubits

(C) 16 rotations, 10 controlled $e^{\pm iAt2^j}$, 8 qubits

(D) 31 rotations, 5 controlled $e^{\pm iAt2^j}$, 8 qubits

Answer: (B) Detailed solution

Correct answer. (B) 31 rotations, 10 controlled $e^{\pm iAt2^j}$, 8 qubits
Concepts required. The naive eigenvalue inversion uses one multi-controlled rotation per non-zero clock value.
Relevant formulae. rotations $= 2^{n_c} - 1$; controlled evolutions $= 2n_c$; qubits $= n_c + n_b + 1$.
Step-by-step explanation. Rotations: one per non-zero clock value, $2^{n_c} - 1 = 31$ (exponential in n_c — this naive inversion does not scale; practical versions compute $1/\lambda$ arithmetically or use QSVT). Controlled evolutions: n_c in phase estimation plus n_c in uncomputation, $2 \times 5 = 10$. Qubits: $5 + 2 + 1 = 8$. (Each rotation is now controlled by 5 qubits.)
Why the other options are incorrect.
(A) 7 qubits is the $n_c = 4$ count; one more clock qubit means 8.
(C) $16 = 2^4$ is the old count $+1$; it should be $2^5 - 1$.
(D) 5 counts only the QPE half; uncomputation needs another 5 inverse evolutions.
Key takeaway. The look-up-table inversion is exponential in n_c: fine for demonstrations, not for scaling.

Q.74 MATCHING LISTS · Error sources and their effects

Match each error source (List-I) with its main effect on HHL (List-II).

List-I		List-II	
(P)	bit flip on the ancilla just before measurement	**(1)**	statistical error $\sim 1/\sqrt{N_{\text{acc}}}$ in the estimated $\lvert x_i\rvert^2$
(Q)	phase error on a clock qubit during phase estimation	**(2)**	wrong eigenvalue estimate, so a wrong factor $1/\tilde{\lambda}$ is applied
(R)	imperfect uncomputation of the clock	**(3)**	failed runs are accepted (and good ones rejected): biased P_{success} and contaminated output
(S)	finite number of shots	**(4)**	residual clock–system entanglement: the output is a mixed state

(A) P→3, Q→2, R→1, S→4
(B) P→2, Q→3, R→4, S→1
(C) P→3, Q→4, R→2, S→1
(D) ✓ P→3, Q→2, R→4, S→1

Answer: (D) Detailed solution

Correct answer. (D) P→3, Q→2, R→4, S→1
Concepts required. Each part of HHL has a characteristic failure mode.
Relevant formulae. ancilla flip: wrong branch; clock error: $1/\tilde{\lambda} \neq 1/\lambda$; residual entanglement: mixed ρ; shots: $\sigma \propto 1/\sqrt{N_{\text{acc}}}$.
Step-by-step explanation.

- P: the ancilla is the success flag; flipping it mixes the $\sqrt{1 - C^2/\lambda^2}$ branch (not $\propto A^{-1}$) into the accepted data (3).
- Q: QPE reads phases; a phase error shifts the estimated k, and the rotation then uses the wrong $\tilde{\lambda}$ (2).
- R: if $|k_j\rangle$ is not reset to $|0\rangle$, the eigencomponents stay labelled and do not interfere: mixed output (4).
- S: sampling noise with the *accepted* shots as sample size (1).

HHL's deep circuits (controlled e^{iAt2^j} with large 2^j) make it far beyond today's noisy hardware for non-trivial sizes.

Why the other options are incorrect.

(A) R→1, S→4: finite shots give statistical error, not entanglement.

(B) P→2, Q→3 swaps the ancilla and clock errors: phase errors in QPE corrupt eigenvalue estimates.

(C) Q→4, R→2 swaps clock errors and imperfect uncomputation.

Key takeaway. Debugging HHL: check eigenvalue registers, the success flag and the uncomputation separately.

Q.75 SINGLE CORRECT · Useful outputs of HHL

Which output can HHL provide efficiently (under its assumptions)?

(A) The full inverse matrix A^{-1}

(B) ✓ An estimate of an expectation value $\langle x|M|x\rangle$ for an efficiently measurable observable M

(C) The determinant of A

(D) The complete vector $\vec{x}$ to 16 significant digits

Answer: (B) Detailed solution

Correct answer. (B) An estimate of an expectation value $\langle x|M|x\rangle$ for an efficiently measurable observable M

Concepts required. HHL is efficient for estimating scalar properties of $|x\rangle$, not for outputting vectors or matrices.

Relevant formulae. $\langle x|M|x\rangle$ to precision ϵ: $O(1/\epsilon^2)$ samples ($O(1/\epsilon)$ with amplitude estimation).

Step-by-step explanation. Measuring an observable M on $|x\rangle$ repeatedly estimates $\langle x|M|x\rangle$ to precision ϵ with $O(1/\epsilon^2)$ runs (or $O(1/\epsilon)$ with amplitude estimation), independently of d. Examples: the total weight of a region of $\vec{x}$, an overlap $|\langle y|x\rangle|^2$ via a swap test, a cost function in machine learning. Printing $\vec{x}$ needs $\Omega(d)$ runs (Q.64); $\det A$ and A^{-1} (d^2 numbers) are not produced by HHL at all.

Why the other options are incorrect.

(A) A^{-1} has d^2 entries; HHL only applies A^{-1} to one state.

(C) HHL does not compute determinants.

(D) The complete vector needs $\Omega(d)$ runs (Q.64).

Key takeaway. Ask HHL scalar questions about $\vec{x}$.

Q.76 SINGLE CORRECT · Dequantisation

"Dequantisation" results (Tang 2019 and successors) give classical algorithms with run time polylogarithmic in d for linear systems when

(A) A is Hermitian

(B) ✓ A has low rank and ℓ^2-norm sample-and-query access to A and $\vec{b}$ is available

(C) $\vec{b}$ is a computational basis vector

(D) A is sparse and well conditioned with $\kappa = \text{polylog}(d)$, and $\vec{b}$ is given as an efficient circuit

Answer: (B) Detailed solution

Correct answer. **(B)** A has low rank and ℓ^2-norm sample-and-query access to A and $\vec{b}$ is available
Concepts required. Quantum-inspired classical algorithms use sample-and-query access (the classical analogue of QRAM) and exploit low rank.
Relevant formulae. classical time $\text{poly}(\text{rank}, \kappa, 1/\epsilon) \cdot \text{polylog}(d)$ for low-rank A with ℓ^2 sampling access.
Step-by-step explanation. Tang's quantum-inspired algorithms show that if one can sample rows/entries with probability proportional to their squared norms (the classical analogue of amplitude encoding / QRAM), low-rank problems such as recommendation systems and low-rank linear systems can be solved classically in time $\text{poly}(\text{rank}, \kappa, 1/\epsilon)\,\text{polylog}(d)$. For sparse, well-conditioned, full-rank systems with efficiently prepared $|b\rangle$, HHL-type problems are BQP-complete and no such classical algorithm is expected. Hermiticity or a basis vector $\vec{b}$ alone gives no classical shortcut.
Why the other options are incorrect.

(A) Hermiticity alone gives no classical shortcut.

(C) A basis-vector $\vec{b}$ does not make $A^{-1}\vec{b}$ classically easy for a general A.

(D) Sparse, well-conditioned, full-rank systems with efficient $|b\rangle$ are BQP-complete; dequantisation is not expected there.

Key takeaway. Low rank + sampling access = classical polylog; sparse full-rank + small κ = where quantum advantage may remain.

Q.77 ASSERTION–REASON · κ^2 run-time dependence

Assertion (A): At fixed N, s and ϵ, doubling the condition number roughly quadruples the run time of the original HHL algorithm.

Reason (R): The original HHL run time scales as $\tilde{O}(\log(N)s^2\kappa^2/\epsilon)$.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
Concepts required. With other parameters fixed, HHL's run time is quadratic in κ.
Relevant formulae. $T \propto \kappa^2 \Rightarrow T(2\kappa)/T(\kappa) = 4$.
Step-by-step explanation. **R is true** (Q.61). **A is true and follows from R**: with everything else fixed, $T \propto \kappa^2$, so $\kappa \to 2\kappa$ gives $T \to 4T$ (up to logarithmic factors). The two factors of κ come from phase-estimation precision ($t \sim \kappa/\epsilon$) and from the repetitions after amplitude amplification ($\sim \kappa$). Improved algorithms (Q.63) reduce this to linear. Answer: (A).
Why the other options are incorrect.

(B) denies the explanatory link, but A is computed directly from R.

(C) rejects R, which is the original HHL bound.

(D) rejects A, but κ^2 scaling gives exactly a factor 4.

Key takeaway. κ is the price tag of HHL; improved algorithms make it linear.

Q.78 SINGLE CORRECT · Solution inside the embedded system

For a non-Hermitian invertible A the notebook solves $\begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}\begin{pmatrix} \vec{y} \\ \vec{z} \end{pmatrix} = \begin{pmatrix} \vec{b} \\ \vec{0} \end{pmatrix}$. The solution satisfies

(A) $\vec{y} = A^{-1}\vec{b}$ (first half) and $\vec{z} = \vec{0}$

(B) $\vec{y} = \vec{z} = A^{-1}\vec{b}/\sqrt{2}$

(C) $\vec{y} = A^{-1}\vec{b}$ and $\vec{z} = (A^\dagger)^{-1}\vec{b}$

(D) ✓ $\vec{z} = A^{-1}\vec{b}$ (second half) and $\vec{y} = \vec{0}$

Answer: (D) Detailed solution

Correct answer. (D) $\vec{z} = A^{-1}\vec{b}$ (second half) and $\vec{y} = \vec{0}$

Concepts required. The Hermitian embedding solves the original system in one block of a doubled system.

Relevant formulae. $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} \vec{y} \\ \vec{z} \end{smallmatrix}\right) = \left(\begin{smallmatrix} A\vec{z} \\ A^\dagger\vec{y} \end{smallmatrix}\right)$.

Step-by-step explanation. Multiply out: the first block row gives $A\vec{z} = \vec{b}$, so $\vec{z} = A^{-1}\vec{b} = \vec{x}$; the second gives $A^\dagger\vec{y} = \vec{0}$, so $\vec{y} = \vec{0}$ ($A^\dagger$ is invertible). Hence the solution sits in the *second* half of the doubled system register (code: `solution_slice = slice(d0, 2 * d0)`). The extra qubit distinguishes the halves; post-selecting it on 1 isolates $|x\rangle$ (here it is 1 with certainty, since $\vec{y} = \vec{0}$).

Why the other options are incorrect.

(A) The first block row involves $A\vec{z}$, so the solution is $\vec{z}$ (second half), not $\vec{y}$.

(B) The solution is not split over both halves; $\vec{y}$ vanishes exactly.

(C) $A^\dagger\vec{y} = \vec{0}$ forces $\vec{y} = \vec{0}$; neither half contains $(A^\dagger)^{-1}\vec{b}$.

Key takeaway. Embedding costs one qubit; the answer is in the half selected by that qubit.

Paragraph for Questions 79 and 80

The previously recorded default run of `HHL.ipynb` (7 qubits, `qpp-cpu`) printed:

```
Success probability:  state 0.355903, shots 0.3543 (7086 of 20000), ideal 0.355903
Fidelity |<x_classical|x_quantum>|^2 :  1.0000000000
Relative error of the recovered vector:  1.655e-15
```

In the eigenbasis of the default A (eigenvalues $1, 2, 3, 4$) the vector $|b\rangle = |00\rangle$ has all four coefficients $\beta_j = \frac{1}{2}$; $C = 1$ and $\|\vec{b}\| = 1$.

Q.79 PARAGRAPH BASED · Recorded run: P_{success} exactly

The exact value of the ideal success probability 0.355903 is

(A) ✓ $\frac{205}{576}$ **(B)** $\frac{25}{48}$ **(C)** $\frac{205}{144}$ **(D)** $\frac{1}{4}$

Answer: (A) Detailed solution

Correct answer. (A) $\frac{205}{576}$

Concepts required. Exact success probability from the eigenbasis coefficients and the eigenvalues.

Relevant formulae. $P = \sum_j |\beta_j|^2 C^2/\lambda_j^2$; $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} = \frac{205}{144}$.

Step-by-step explanation. $P_{\text{success}} = \sum_j |\beta_j|^2 \frac{C^2}{\lambda_j^2} = \frac{1}{4}\left(1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16}\right) = \frac{1}{4} \cdot \frac{144+36+16+9}{144} = \frac{205}{576} =$ 0.3559028. It also equals $C^2\|\vec{x}\|^2/\|\vec{b}\|^2 = \frac{25^2+7^2+11^2+5^2}{48^2} = \frac{820}{2304} = \frac{205}{576}$. ($\frac{205}{144}$ forgets $|\beta_j|^2 = \frac{1}{4}$; $\frac{25}{48}$ is x_0.) Since every λ_j is exactly representable, the state-vector run reproduces it to all printed digits.

Why the other options are incorrect.

(B) $\frac{25}{48}$ is the first solution component x_0.

(C) $\frac{205}{144} > 1$ forgets $|\beta_j|^2 = \frac{1}{4}$ — impossible for a probability.

(D) $\frac{1}{4} = |\beta_j|^2$ for one component; it ignores the $1/\lambda_j^2$ factors.

Key takeaway. $P_{\text{success}} = C^2\|\vec{x}\|^2/\|\vec{b}\|^2$: here $\frac{205}{576}$ exactly.

Q.80 PARAGRAPH BASED · Recorded run: $\|\vec{x}\|$ from P_{success}

Without knowing $\vec{x}$ in advance, the norm $\|\vec{x}\|$ can be inferred from the success probability. Its value is

(A) $\frac{24}{\sqrt{205}} \approx 1.676$ **(B)** $\frac{205}{576} \approx 0.356$

(C) ✓ $\frac{\sqrt{205}}{24} \approx 0.597$ **(D)** 1, because quantum states are normalised

Answer: (C) Detailed solution

Correct answer. (C) $\frac{\sqrt{205}}{24} \approx 0.597$

Concepts required. The success probability measures the norm of the post-selected branch, from which $\|\vec{x}\|$ follows.

Relevant formulae. $\|\vec{x}\| = \frac{\|\vec{b}\|}{C}\sqrt{P_{\text{success}}}$.

Step-by-step explanation. The accepted branch is $\vec{a} = \frac{C}{\|\vec{b}\|}\vec{x}$ (Q.53) and $\|\vec{a}\|^2 = P_{\text{success}}$, hence

$$\|\vec{x}\| = \frac{\|\vec{b}\|}{C}\sqrt{P_{\text{success}}} = \sqrt{\frac{205}{576}} = \frac{\sqrt{205}}{24} \approx 0.5966.$$

Check: $\frac{1}{48}\sqrt{820} = \frac{2\sqrt{205}}{48} = \frac{\sqrt{205}}{24}$. On hardware the measured success rate (0.3543 ± 0.0034) gives $\|\vec{x}\| \approx 0.595 \pm 0.003$. The normalised state $|x\rangle$ has norm 1, but the solution vector does not.

Why the other options are incorrect.

(A) $\frac{24}{\sqrt{205}}$ inverts the relation.

(B) $\frac{205}{576}$ is $\|\vec{x}\|^2$; the square root is missing.

(D) 1 is the norm of the normalised state $|x\rangle$, not of $\vec{x}$.

Key takeaway. Measure the success rate to recover the scale of $\vec{x}$ that the normalised state forgets.